

CHEMOTHERAPY

General Principles of Antimicrobial Therapy

Introduction

- Antimicrobials are among the most commonly used and misused of all drugs
- Widespread and irrational use of antimicrobial agents $\Rightarrow$ Emergence of antibiotic-resistance

Definition

➤ Antibiotics

- are antibacterial substances produced by various species of microorganisms that suppress the growth of other microorganisms.

➤ Commonly the term extends to include synthetic antimicrobial agents, such as *sulfonamides* and *quinolones*.

Definitions...

- **Chemotherapy**

- *Use of chemicals (drugs) against invading organisms as well as cancerous cells.*

- **Antimicrobial Agent**

- *Use of chemicals against invading organisms.*

- **Antibiotic**

- *is a chemical that is produced by one microorganism and has the ability to harm other microbes.*

Definition

- **Bactericidal**

Is directly lethal to bacterial at clinically achievable concentrations

- **Bacteriostatic**

seize microbial growth but does not cause cell death

- elimination of bacteria must ultimately be accomplished by host defenses

Definition

- **Minimal inhibitory concentration (MIC):** Lowest concentration of antimicrobial drug capable of inhibiting growth of an organism in a defined growth medium
- **Selective toxicity:** More toxic to the invader than to the host; a property of useful antimicrobial drugs

Selective toxicity

- Is the ability of a drug to harm a target cell or target organism with out injuring other cells or organisms with which the target is in intimate contact.
- Antimicrobial drugs kill or suppress infecting microbes with out causing injury to the host.
- If they were as harmful to the host as they are to infecting organisms, antimicrobials would have no therapeutic utility.

Classification

- Based on chemical structure and proposed MOA
 1. Cell wall synthesis inhibitors , *e.g.* β -lactam antibiotics, *cycloserine*, *vancomycin*, and *bacitracin*
 2. Agents that act directly on the cell membrane of the microorganism, *e.g.* *Polymyxin*, *nystatin* and *amphotericin B*, daptomycin
 3. Agents that affect ribosomal protein synthesis (*e.g.*, *CAF*, *TTCs*, *erythromycin*, *clindamycin*, *aminoglycosides*)
 4. Agents that affect bacterial nucleic acid metabolism, such as the rifamycins and the quinolones
 5. The antimetabolites, including *trimethoprim* & *sulfonamides*

Susceptibility and Resistance

- Antimicrobial therapy depends on the concentration of antibiotic at the site of infection.
 - Concentration should be sufficient to inhibit growth
 - Intact and active host defence $\Rightarrow$ *bacteriostatic* effect sufficient
 - Impaired host defence $\Rightarrow$ antibiotic-mediated killing required
 - The concentration at the site of infection
 - Inhibit the organism + remain below the level that is toxic to humans
 - Susceptible
 - If an inhibitory or bactericidal concentration exceeds that which can be achieved safely *in vivo*, then the microorganism is considered resistant to that drug.

General mechanisms of bacterial resistance

- The drug does not reach its target
- The drug is not active
- The target is altered

- Drug resistance may be acquired by
- **Mutation and selection**, with passage of the trait *vertically* to daughter cells
 - *Horizontal transfer* of resistance determinants from a donor cell, often of another bacterial species, by **transduction, transformation, or conjugation.**

➤ Antibiotics have three general uses

- Empirical (initial) therapy
 - ✓ Antibiotic should cover all the likely pathogens
 - ✓ Either combination therapy or a single broad-spectrum agent
- Definitive therapy
 - ✓ Once the infecting microorganism is identified, a narrow-spectrum, low-toxicity agent is instituted
- Prophylactic or preventive therapy

Factors that affect selection of antibiotics

- Microbial sensitivity
- Pharmacokinetic factors
- *Route of Administration*
- *Host Defense Mechanisms*
- *Local Factors*
- *Age*
- *Pregnancy*
- *Drug allergy*

Combining Antimicrobial Agents

- Recommended in specifically defined situations based on pharmacological rationale.
- **Selection of an appropriate combination**
 - Understanding of the potential for interaction between the antimicrobial agents.
 - Interactions may affect either the microorganism or the patient.
 - May enhance or impair overall antimicrobial activity.
 - May have additive or superadditive toxicities.

➤ **Indications for the Clinical Use of Combinations of Antimicrobial Agents**

- For empirical therapy of an infection in which the cause is unknown
- For treatment of polymicrobial infections
- To enhance antimicrobial activity (*i.e.* synergism) for a specific infection
- To prevent emergence of resistance.

Disadvantages of Combinations of Antimicrobial Agents

- Increased risk of toxicity from two or more agents
- Selection of multiple-drug-resistant microorganisms
- Eradication of normal host flora with subsequent superinfection
- Increased cost to the patient.

Super-infections

- Appearance of bacteriological and clinical evidence of a new infection during the chemotherapy of a primary one
- Alterations in the normal microbial population of the intestinal, upper respiratory, and genitourinary tracts
- Due to removal of inhibitory influence of normal flora
- **The broader the antibacterial spectrum and the longer the period of antibiotic treatment, the greater is the alteration in the normal microflora, and the greater is the possibility that a single, typically drug-resistant microorganism will become predominant, invade the host, and produce infection.**

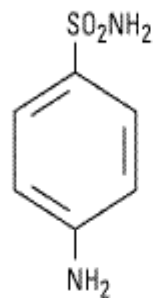
Misuses of Antibiotics

- ☞ Treatment of nonresponsive infections
- ☞ Therapy of fever of unknown origin
- ☞ Improper Dosage
- ☞ Inappropriate reliance on chemotherapy alone
- ☞ Lack of adequate bacteriological information

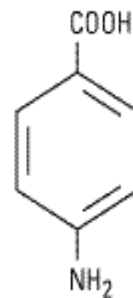
SULFONAMIDES

Chemistry

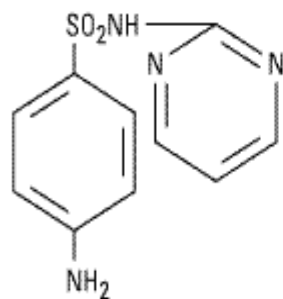
- Has the basic **sulfanilamide** nucleus (which is structurally similar to **p-aminobenzoic acid**)
 - Derivatization: attaching substituents to the **amido** group ($-\text{SO}_2-\text{NH}-\text{R}$) or the **amino** group ($-\text{NH}_2$)
- Sulfonamides tend to be much more soluble at alkaline than at acidic pH.



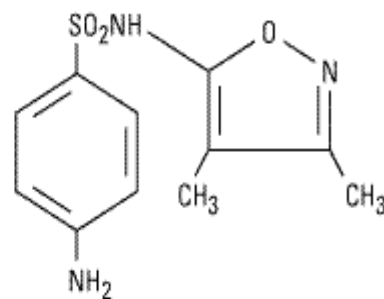
Sulfanilamide



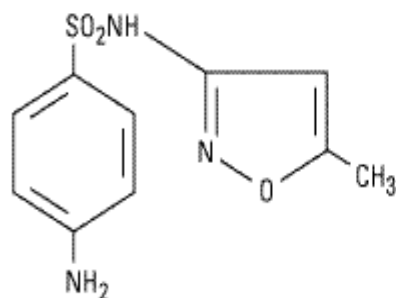
***p*-Aminobenzoic acid (PABA)**



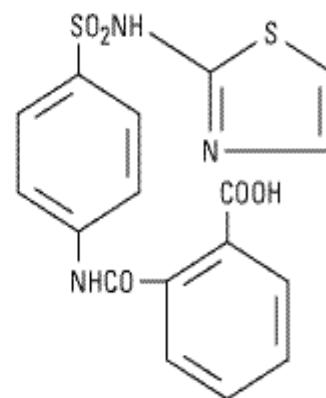
Sulfadiazine



Sulfisoxazole



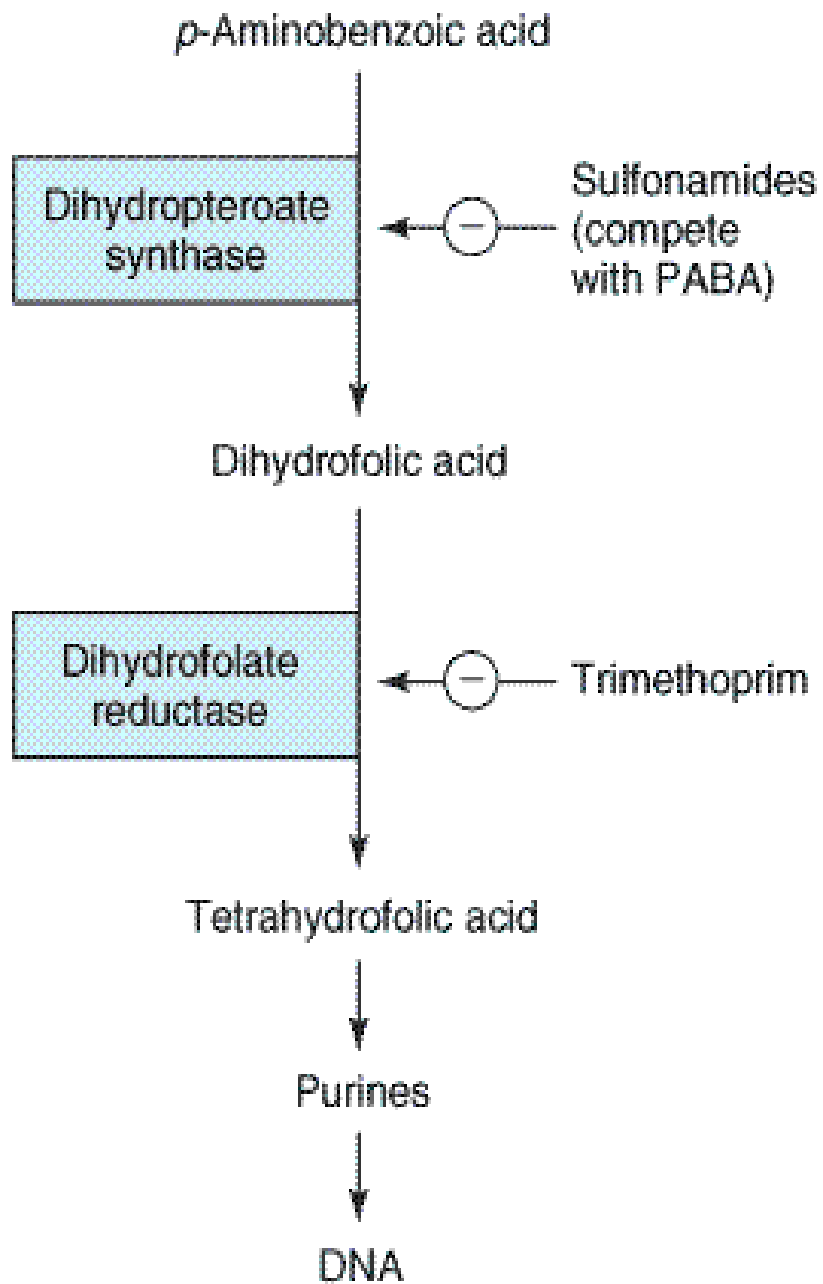
Sulfamethoxazole

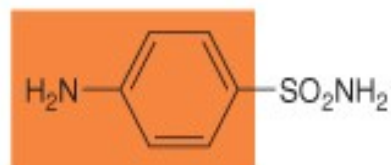
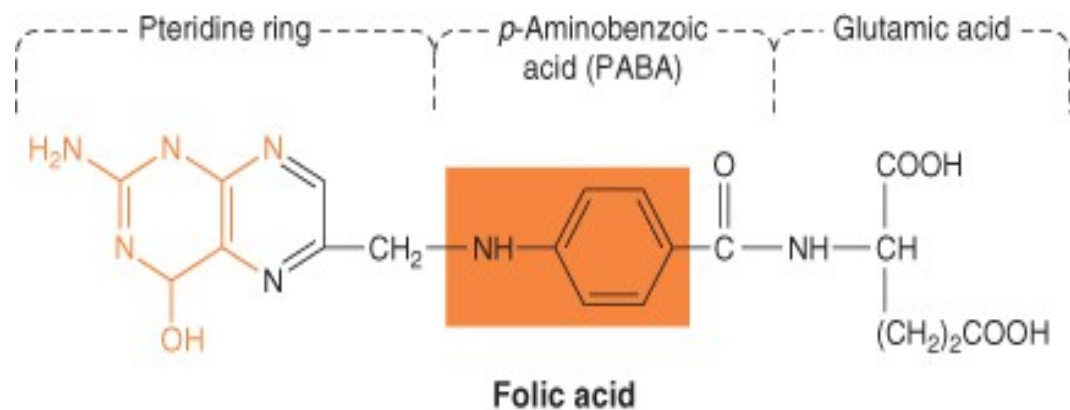


**Sulfathalidine
(phthalylsulfathiazole)**

Mechanism of action and resistance

- Susceptible microorganisms require extracellular PABA to form dihydrofolic acid (*Humans cannot synthesize folic acid and must acquire it in the diet*)
- Competitively inhibit dihydropteroate synthase (DHPS) and block folic acid synthesis
- *bacteriostatic*
- Inhibit both G(+) and G(-) bacteria, nocardia, *Chlamydia trachomatis*, Some enteric bacteria (*E coli, klebsiella, salmonella, shigella, and enterobacter*), and some protozoa.

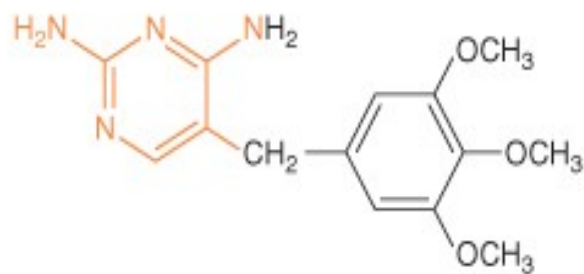




Sulfanilamide



Sulfadiazine



Trimethoprim (dihydrofolate reductase inhibitor)

Mechanism of action and resistance (cont'd)

- Resistance can be the result of
 - Decreased bacterial permeability to the drug
 - **Increased production of PABA**
 - **Production of an altered DHPS that exhibits low affinity for sulfonamides.**
 - Active efflux of the sulfonamides
 - Their inhibitory effect can be reversed by the presence of pus, tissue fluids, and drugs that contain releasable PABA.

Pharmacokinetics

➤ Divided into three major groups

1. Oral, absorbable
 - ✓ Short-, medium-, or long-acting on the basis of their half-lives.
2. Oral, nonabsorbable
3. Topical

Drug	Half-Life	Oral Absorption
Sulfonamides		
Sulfacytine	Short	Prompt (peak levels in 1–4 hours)
Sulfisoxazole	Short (6 hours)	Prompt
Sulfamethizole	Short (9 hours)	Prompt
Sulfadiazine	Intermediate (10–17 hours)	Slow (peak levels in 4–8 hours)
Sulfamethoxazole	Intermediate (10–12 hours)	Slow
Sulfapyridine	No data	Slow
Sulfadoxine	Long (7–9 days)	Intermediate

Clinical Uses

- Infrequently used as single agents.
- largely replaced by trimethoprim-sulfamethoxazole (Co-trimoxazole).
- Many strains of formerly susceptible species are now resistant.
- **Can be useful for treatment of urinary tract infections due to susceptible organisms**

Clinical Uses...

- **Urinary Tract Infections**
- **Nocardiosis**
- **Toxoplasmosis** (combination of *pyrimethamine* and sulfadiazine is the treatment of choice)

Rapidly Absorbed and Eliminated Sulfonamides

➤ *Sulfisoxazole*

- Rapidly absorbed and excreted
- **High solubility minimizes renal toxicity**
- About 95% of a single dose is excreted by the kidney in 24 hours
- Preferred when a rapidly absorbed and rapidly excreted sulfonamide is indicated.

➤ *Sulfamethoxazole*

- A close congener of sulfisoxazole with slower rates of enteric absorption and urinary excretion
- **Crystalluria may occur (high %age of the acetylated, relatively insoluble form in the urine).**
- A component of the fixed-dose combination cotrimoxazole (with trimethoprim)

➤ *Sulfadiazine*

- Sulfadiazine given orally is absorbed rapidly from the GI tract, and excreted quite readily by the kidney in both the free and acetylated forms
- About 15% to 40% of the excreted sulfadiazine is in acetylated form.
- Risk of crystal (ensure adequate fluid intake or sodium bicarbonate)

Poorly Absorbed Sulfonamides

➤ Sulfasalazine

- Used in ulcerative colitis and regional enteritis
- Broken down by intestinal bacteria to *sulfapyridine* and *5-aminosalicylate*
- 5-Aminosalicylate effective in inflammatory bowel disease, whereas sulfapyridine is responsible for most of the toxicity.
- Toxic reactions include acute hemolysis in patients with glucose-6-phosphate dehydrogenase deficiency, and agranulocytosis. Nausea, fever.

Sulfonamides for Topical Use

➤ *Sulfacetamide*

- N₁-acetyl-substituted derivative of *sulfanilamide*.
- Sodium salt solutions are used in ophthalmic infections.
 - ✓ Very high aqueous concentrations are not irritating to the eye and are effective against susceptible microorganisms.
 - ✓ The drug penetrates into ocular fluids and tissues in high concentration.

➤ *Silver Sulfadiazine*

- Used topically to reduce microbial colonization and the incidence of infections of wounds from burns
- It should not be used to treat an established deep infection.
- Silver is released slowly from the preparation in concentrations that are selectively toxic to the microorganisms.
- **One of the agents of choice for the prevention of burn infection.**

Long-Acting Sulfonamides

➤ *Sulfadoxine*

- Has long half-life (7 to 9 days)
- Used in combination with *pyrimethamine* (500 mg sulfadoxine plus 25 mg pyrimethamine as FANSIDAR) for the prophylaxis and treatment of malaria caused by *mefloquine*-resistant strains of *Plasmodium falciparum*

➤ **Adverse Reactions**

- All sulfonamides and their derivatives are cross-allergenic.
- Skin rashes, urticaria, photosensitivity, arthritis, conjunctivitis, hematopoietic disturbances, hepatitis
- Precipitate in urine (at neutral or acid pH) producing crystalluria, hematuria, or even obstruction.

Trimethoprim

- Structural analogue of the pteridine portion of DHF acid
- Competitively inhibits dihydrofolate reductase (DHFR)
- The bacterial enzyme is 20,000 to 60,000 times more sensitive than its mammalian enzyme
- Trimethoprim–sulfamethoxazole
 - Synergistically and sequentially inhibit bacterial synthesis of THF acid
 - To delay development of bacterial resistance

Antibacterial Spectrum & Resistance

- Trimethoprim exhibits broad-spectrum activity
- Active against most g(+) and g(-)
- Little activity against *anaerobic bacteria*
- Resistance can develop from
 - Alterations in DHFR
 - Bacterial impermeability to the drug
 - Overproduction of DHFR

Pharmacokinetics

- Usually given orally, alone or in combination with sulfamethoxazole
- Absorbed efficiently from the gut and distributed widely in body fluids and tissues, including CSF
- Trimethoprim (a weak base of pKa 7.2) concentrates in prostatic and vaginal fluids, which are more acid than plasma.

Clinical use

- In acute urinary tract infections(100 mg twice daily)
- Most community-acquired organisms tend to be susceptible to the high concentrations that are found in the urine.

Trimethoprim 80mg-Sulfamethoxazole 400 mg (TMP-SMX)

- *The combination shows synergistic effect as the two affect different points in the folic acid synthetic pathway*
- The incidence of bacterial resistance is less than that observed when the drugs are used individually.

➤ Oral preparation

- Effective for *P. jiroveci pneumonia*, shigellosis, systemic salmonella infections, complicated UTI, prostatitis, many respiratory tract pathogens (including the pneumococcus, haemophilus species, *Moraxella catarrhalis*, and *Klebsiella pneumoniae*)
 - ✓ *Useful alternative to β -lactams in URTI and community-acquired bacterial pneumonia*

➤ Parenteral preparations

- Agent of choice for moderately severe to severe pneumocystis pneumonia.
- If patient can not take the oral preparation

Pyrimethamine

- 2,4- diaminopyrimidine derivative that inhibits DHFR
- Pyrimethamine is well absorbed after oral administration
- Binds to tissues and has slow rate of renal excretion
- In combination with a sulphonamide
 - Malaria
 - Toxoplasmosis

Adverse Effects

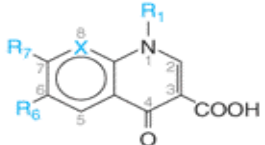
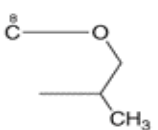
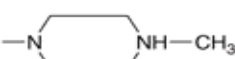
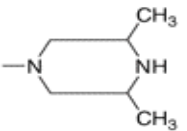
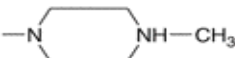
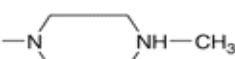
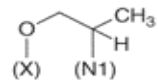
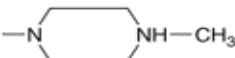
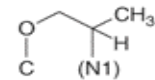
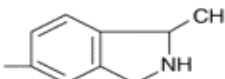
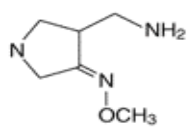
- TMP: antifolate adverse effects (megaloblastic anemia, leukopenia, and granulocytopenia)
 - Prevented by simultaneous administration of folinic acid
- Nausea and vomiting, drug fever, vasculitis, renal damage

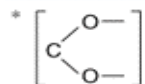
QUINOLONES

Introduction

- Nalidixic acid: the first quinolone
 - Used for many years in treatment of UTIs
- Fluorinated 4-quinolones
 - Broad antimicrobial activity
 - Effective after oral administration for the treatment of a wide variety of infectious diseases
 - Relatively few side effects
 - Microbial resistance does not develop rapidly

Table 43-2. Structural Formulas of Selected Quinolones and Fluoroquinolones

				
CONGENER	R ₁	R ₆	R ₇	X
Nalidixic acid	—C ₂ H ₅	—H	—CH ₃	—N—
Cinoxacin (<i>N</i> replaces C2)	—C ₂ H ₅	(Fused dioxolo ring)*		—CH—
Norfloxacin	—C ₂ H ₅	—F		—CH—
Ciprofloxacin		—F		—CH—
Ofloxacin		—F		—CH—
Sparfloxacin (—NH ₂ on C5)		—F		
Fleroxacin	—CH ₂ —CH ₂ —F	—F		
Pefloxacin	—C ₂ H ₅	—F		—CH—
Levofloxacin		—F		
Garenoxacin		—H		
Gemifloxacin		—F		—N—



MOA

- The quinolone antibiotics target bacterial DNA gyrase and topoisomerase IV
- DNA replication or transcription $\Rightarrow$ double-helical DNA separation $\Rightarrow$ **overwinding or excessive positive supercoiling**
- DNA gyrase introduces negative supercoils
 - An ATP-dependent reaction requiring that both strands of the DNA be cut to permit passage of a segment of DNA through the break; the break then is resealed.

- Topoisomerase IV separates interlinked (catenated) daughter DNA molecules that are the product of DNA replication.
- Eukaryotic cells
 - Do not contain DNA gyrase
 - Have type II DNA topoisomerase
- Quinolones inhibit eukaryotic type II topoisomerase only at much higher concentrations

Classification of quinolones

- Often classified into generations (1st through 4th) with spectrum of specificity and unique pharmacological properties
 - First: nalidixic acid and cinoxacin;
 - Second: **norfloxacin**, **ciprofloxacin**, ofloxacin, enoxacin, and lomefloxacin;
 - Third: levofloxacin, sparfloxacin, gatifloxacin;
 - Fourth: trovafloxacin and moxifloxacin

➤ First-generation

- Oldest quinolones
- Exhibit limited Gm(-) activity
- Nalidixic acid and cinoxacin restricted to therapy of bladder infections caused by urinary pathogens, such as *E. coli* and *Klebsiella* and *Proteus spp.*
- Use is restricted by resistance.

➤ Second-generation

- Demonstrate activity against Gm(-) organisms including Enterobacteriaceae. *Haemophilus spp.* and STD agents are also susceptible.
- *The* piperazine moiety is responsible for the antipseudomonal activity of some of these drugs
- Resistance is becoming more prevalent.

- The third and fourth generations possess significantly greater activity against Gm(+), such as *S. pneumoniae*
- The fourth-generation quinolones also possess activity against anaerobes.

Pharmacokinetics

- Fluoroquinolones are well absorbed (bioavailability of 80–95%) After oral administration
- Distributed widely in body fluids and tissues.
- Oral absorption is impaired by divalent cations
- Most are eliminated by renal mechanisms (tubular secretion or GF. Dose adjustment required in renal impairment.

Mechanism of Resistance

- *Via* mutations in the bacterial chromosomal genes encoding DNA gyrase or topoisomerase IV or by active transport of the drug out of the bacteria.
- No quinolone-modifying or -inactivating activities have been identified in bacteria
- Resistance has increased in *Pseudomonas*, staphylococci, *Salmonella*, *N. gonorrhoeae*, and *S. pneumoniae*

Clinical Uses

- UTI by even MDR bacteria, eg, *Pseudomonas*.
 - Norfloxacin, ciprofloxacin, and ofloxacin
 - Also for bacterial diarrhea caused by shigella, salmonella, *E coli*, or *campylobacter*.
- Infections of soft tissues, bones, and joints and in intra-abdominal and RTI
 - *All fluoroquinolones except norfloxacin,*

Clinical Uses (cont'd)

- Gonococcal infection (Ciprofloxacin and ofloxacin)
- Chlamydial urethritis or cervicitis (ofloxacin).
- Occasionally in the treatment of TB (Ciprofloxacin or levofloxacin)

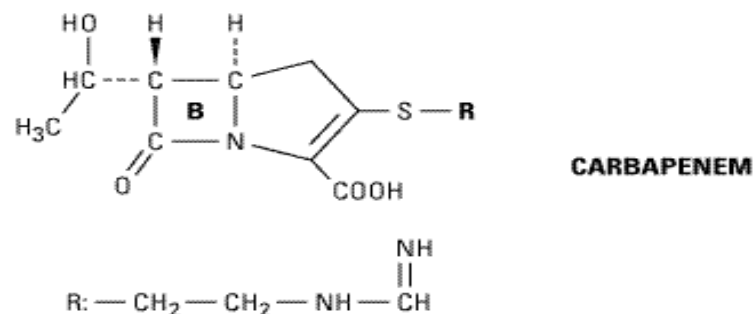
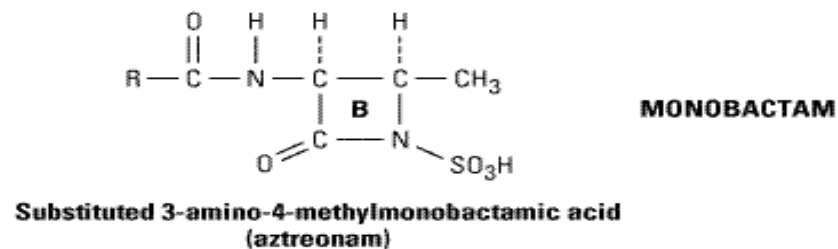
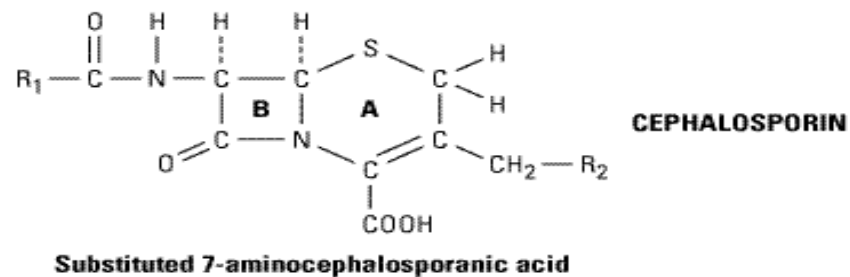
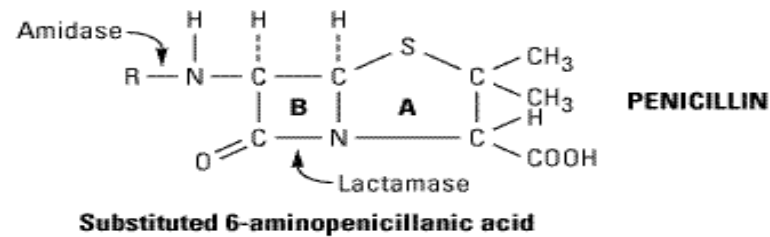
Adverse Effects

- Extremely well tolerated.
- Most common: nausea, vomiting, and diarrhea
- Occasionally, headache, dizziness, insomnia, skin rash, or abnormal liver function tests develop.
- Acute hepatitis and hepatic failure (Trovafloracin)
- Photosensitivity (lomefloracin and pefloracin)
- QT prolongation
- Arthropathy
- They should be avoided during pregnancy in the absence of specific data documenting their safety.

β -LACTAM ANTIBIOTICS

Introduction

- The drugs share a common structure and mechanism of action, inhibition of synthesis of the bacterial peptidoglycan cell wall.
- Include **penicillins, cephalosporins, β -lactamase inhibitors, carbapenems and monobactam.**
- Mechanisms of resistance against β -lactam antibiotics
 - Production of β -lactamases
 - Alterations in or acquisition of novel **penicillin-binding proteins**
 - Decreased entry and/or active efflux of the antibiotic

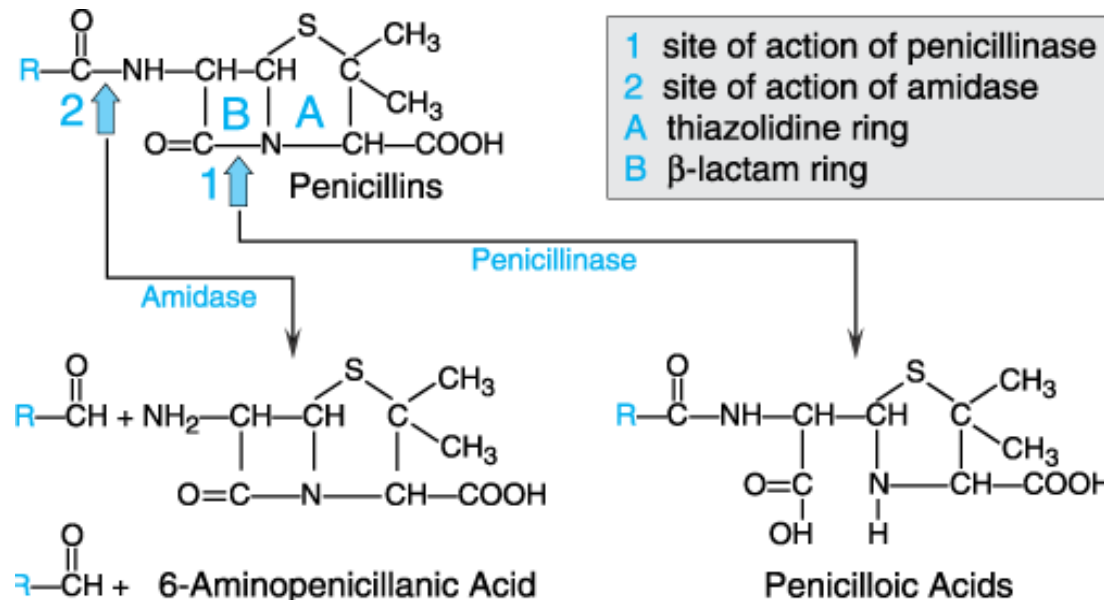


Substituted 3-hydroxyethylcarbapenemic acid (imipenem)

PENICILLINS

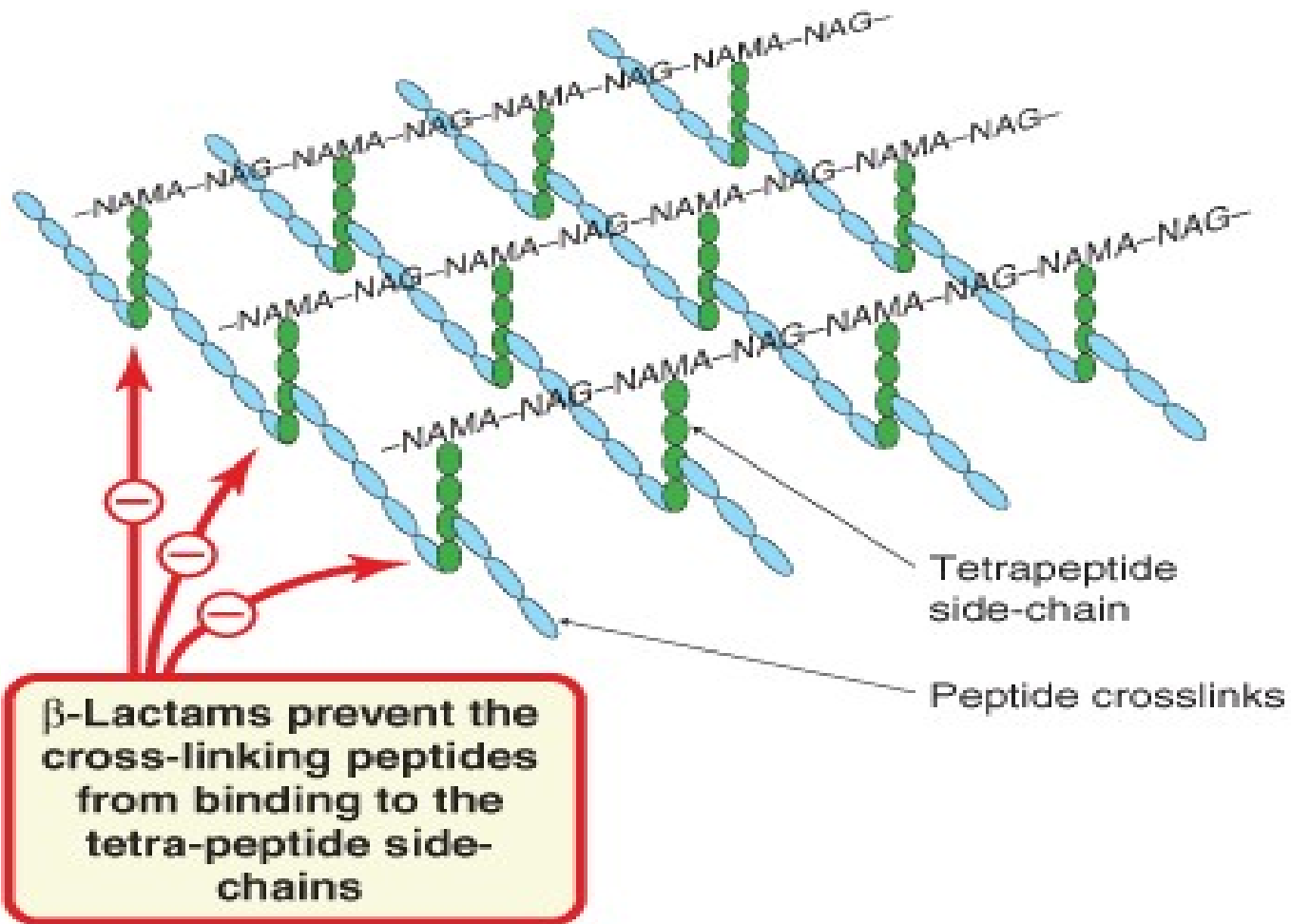
➤ Chemistry

- The basic structure of the penicillins consists
 - ✓ A **thiazolidine ring (A)** connected to
 - ✓ A **β-lactam ring (B)** to which is attached a **side chain (R)**



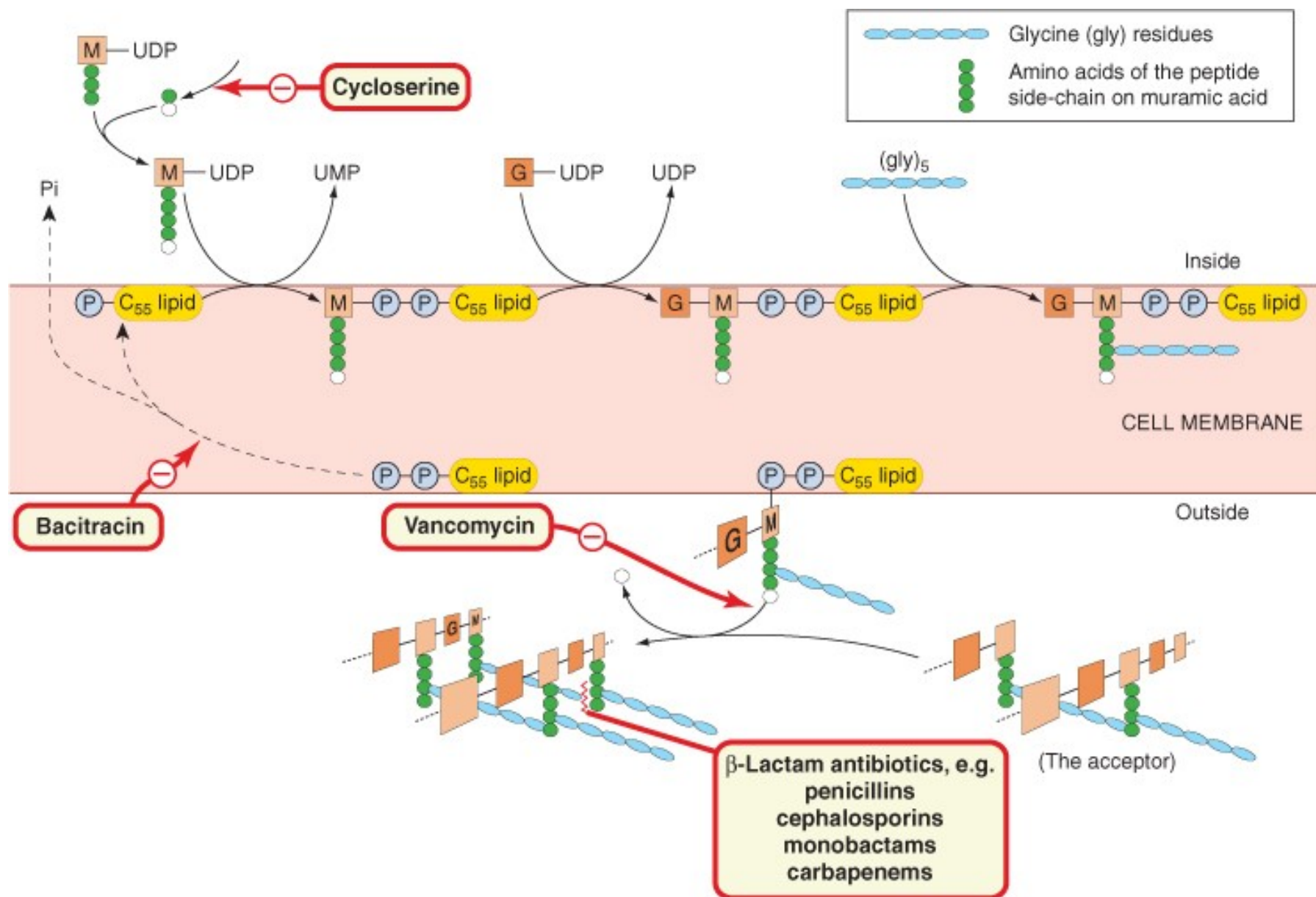
Mechanism of action

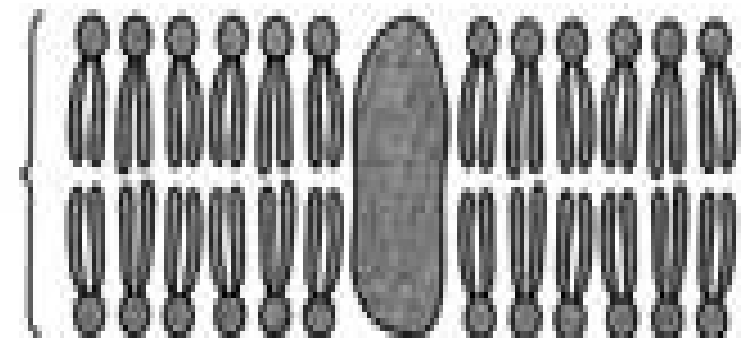
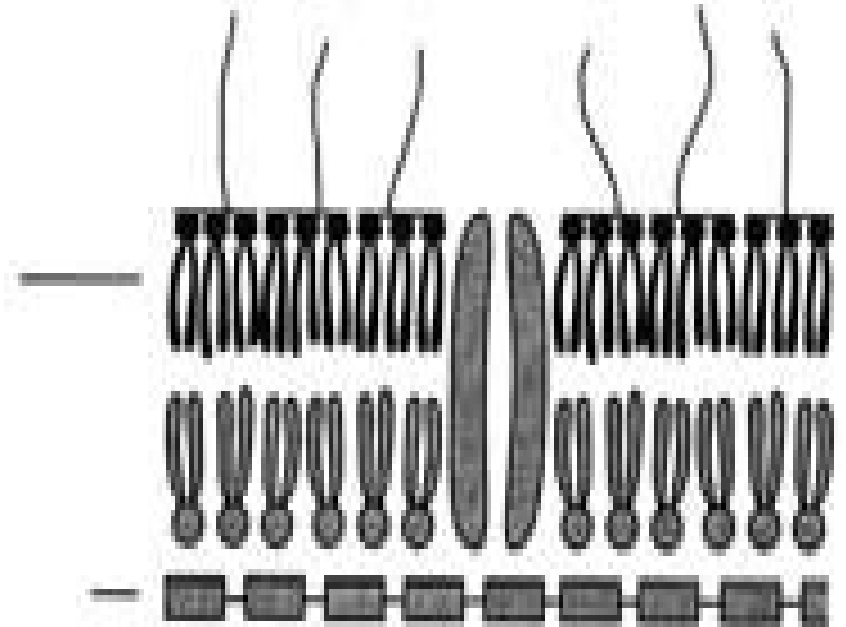
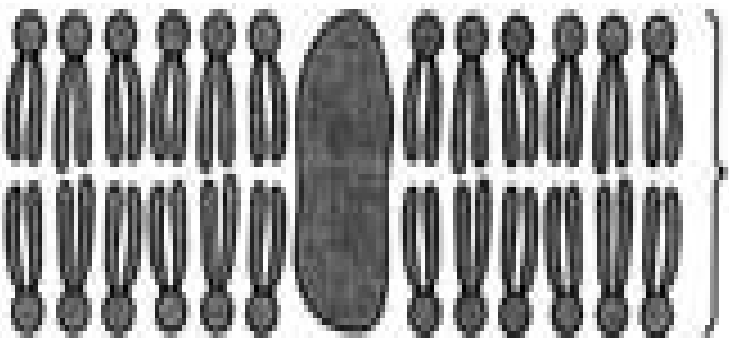
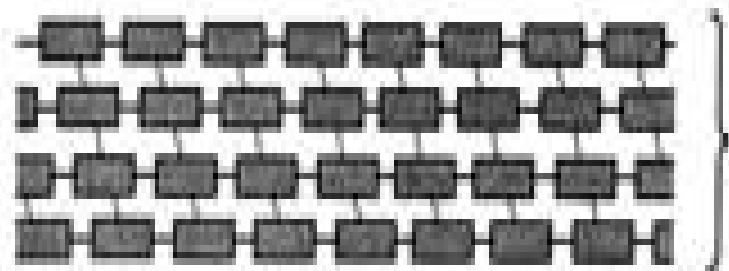
- The β -lactam antibiotics structurally resemble the terminal **D-alanyl-D-alanine** (D-Ala-D-Ala) in the pentapeptides on peptidoglycan (murein).
- Bacterial **transpeptidases** covalently bind β -lactam antibiotics at the enzyme active site, and the resultant acyl enzyme molecule is stable and inactive.
- The β -lactam ring modifies the active serine site on transpeptidases and blocks further enzyme function.



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- In addition, penicillins bind to penicillin-binding proteins (PBPs), which function as **transglycosylases** and **carboxypeptidases**.
- PBPs are involved with assembly, maintenance, or regulation of peptidoglycan cell wall synthesis.





Classification

- Currently penicillins are divided into 4 groups based on the spectra of activity
 - Natural penicillins
 - ✓ Penicillin G, Penicillin V
 - Penicillinase-resistant penicillins
 - ✓ Dcloxacillin, Nafcillin, Oxacillin
 - Aminopenicillins
 - ✓ Amoxicillin, Ampicillin
 - Extended-spectrum penicillins
 - ✓ Carbenicillin, Piperacillin, Ticarcillin

Drug	Route	Half-life (hr)	Renal Excretion (%)	Dose Adjustment in Renal Failure
Natural penicillins				
Penicillin G	IM, IV	0.5	79–85	Yes
Penicillin V	Oral	0.5	20–40	Yes
Antistaphylococcal penicillins (penicillinase resistant)				
Nafcillin	IM, IV	0.8–1.2	31–38	No
Oxacillin	IM, IV	0.4–0.7	39–66	No
Cloxacillin	Oral	0.5–0.6	40–70	No
Dicloxacillin	Oral	0.6–0.8	35–90	No
Aminopenicillins				
Ampicillin	Oral, IM, IV	1.1–1.5	40–92	Yes
Amoxicillin	Oral	1.4–2.0	86	Yes
Antipseudomonal penicillins				
Carbenicillin	Oral	0.8–1.2	85	Yes
Mezlocillin	IM, IV	0.9–1.7	61–69	Yes
Piperacillin	IM, IV	0.8–1.1	74–89	Yes
Ticarcillin	IM, IV	1.0–1.4	95	Yes

IM, intramuscular; IV, intravenous.

$$\begin{array}{c} \text{O} \\ || \\ \text{R}-\text{C}-\text{NH}_2-\text{CH}-\text{CH} \quad \text{S} \quad \text{C} \begin{matrix} / \text{CH}_3 \\ \backslash \text{CH}_3 \end{matrix} \\ | \qquad | \qquad | \\ \text{O}=\text{C}-\text{N}-\text{CH}-\text{COOH} \end{array} \quad \text{Pe}$$
c1ccccc1COCOc1ccccc1OCCOc1ccccc1OCR1c1ccc(cc1)CNc1ccccc1C(C(=O)OR)C(=O)ORC1=CC=C(S1)C(=O)Oc1ccccc1CNC(=O)O

A. Natural Penicillins

- Produced by fermentation of *Penicillium chrysogenum*.
- Different side chains at R are produced by altering the composition of the culture media of *Penicillium*
 - Penicillins F, G, N, O, V, X
- Are acid labile and β -lactamase sensitive
- Active against most **Gm(+)** and **Gm(-) aerobic** cocci, some **Gm(+)** **aerobic** and **anaerobic bacilli**, and most **spirochetes**.
 - Many bacteria have developed resistance.
 - Most streptococci are susceptible
 - More than 90% of staphylococcal strains are now resistant

➤ Penicillin G

- Non-suitable for oral administration (IM or IV)
- The drug distributes to most tissues
- **Excreted** by the kidneys (90% of renal elimination occurs via tubular secretion and 10% by glomerular filtration.
- Probenecid blocks tubular secretion and has been used to prolong the half-life of penicillins).
- Depot **IM** formulations of penicillin G (**procaine** penicillin and **benzathine** penicillin) have decreased solubility, delayed absorption, and a prolonged $t_{1/2}$
- **IV** penicillin G is among the antibiotics of first choice in meningitis caused by susceptible *Streptococcus pneumoniae*

➤ Penicillin G **potassium** and Penicillin G **sodium**

- Aqueous, crystalline Pen G and are used IV
- Used when rapid and high serum concentrations are required
 - ✓ *Septicemia, meningitis, pericarditis, endocarditis, severe pneumonia*

➤ Penicillin G procaine and penicillin G benzathine

- Used only
 - ✓ For treatment of **moderately** severe infections susceptible to low concentrations
 - ✓ Prophylaxis of infections caused by these organisms
 - ✓ Or maintenance therapy to IM or IV Penicillin G

➤ Penicillin V

- **Phenoxymethyl** congener of penicillin G
- More **stable** in **acidic** media and is better absorbed from the GIT
- Have similar antibacterial spectrum as penicillin G.
- Used to treat *streptococcal* infections when oral therapy is appropriate and desirable.

➤ Penicillin G Uses

▪ **Gm(+) aerobic bacterial infections**

- ✓ Drugs of choice for many ***streptococcal*** infections and non-penicillinase producing staphylococcal infections
- ✓ Drugs of first or second choice for the treatment of many infections caused by susceptible **Gm(+) aerobic bacilli** (*Bacillus anthracis*, *Corynebacterium diphtheria*..)

▪ **Gm(-) aerobic bacterial infections**

- ✓ Treatment of a variety of infections by susceptible ***cocci*** (*Neisseria meningitidis*)
- ✓ Drugs of choice in infections caused by ***pasteurella***
- ✓ Generally **not useful** in the treatment of infections caused by other Gm(-) aerobic bacteria

➤ Penicillin G Uses (cont'd)

- **Anaerobic bacterial infection**

- ✓ Drugs of choice for the treatment of many infections including those caused by
 - ✓ Actinomyces, Peptococcus, Peptostreptococcus, and some strains of Bacterioides, Clostridia, Eubacterium, and Fusobacterium

B. Penicillinase-resistant penicillins

- Also Known As *antistaphylococcal* penicillins
- Semi-synthetic antibiotics resistant to staphylococcal penicillinase (hence effective against streptococci and most **community-acquired** penicillinase- producing staphylococci)
- Are relatively stable in acidic medium
- Have **bulky side chains** at R on the 6-APA
 - Results in steric hindrance for the binding of penicillinases to the β -lactam ring
- Generally less active against Pen G susceptible organisms than other penicillins

- Methicillin-resistant *S. aureus* (MRSA): emerging
 - Resistance of bacteria to all the penicillinase-resistant penicillins and cephalosporins
 - **Vancomycin** considered drug of choice for MRSA
- Nafcillin and oxacillin
 - Parenteral
- Cloxacillin and dicloxacillin
 - Oral
- Methicillin: no more in clinical use

C. Aminopenicillins

- Semisynthetic penicillins with enhanced activity against Gm(-) bacteria compared to natural and penicillinase-resistant penicillins
- Are stable in acidic media
- Amoxicillin & Ampicillin
 - Used orally for the treatment of infections of upper & lower RTI, GIT, skin and skin structures, genitourinary tract, & otitis media caused by susceptible organisms.
 - IM or IV for the treatment of meningitis, endocarditis, or severe RT, GIT, bone and joint, or genitourinary tract infections by susceptible organisms.

➤ Aminopenicillins are used for the treatment of

- infections caused by susceptible Gm(-) aerobic bacilli

- ✓ *Haemophilus influenzae*, *E. coli*, *Salmonella*

- Susceptible Gm(+) aerobic cocci

- ✓ *Enterococci*, *Streptococcus pneumonia*, non-penicillinase-producing *S. aureus*)

- Gm(+) bacilli

- ✓ *Listeria monocytogenes*

- Amoxicillin is absorbed more rapidly and completely from the GIT than is ampicillin (the major difference between the two drugs)
- Both have essentially identical spectrum of antimicrobial action, except that amoxicillin is less effective than ampicillin for shigellosis.

D. Extended-spectrum penicillins

- Semisynthetic penicillins with wider spectra of activity than the other penicillins.
- Are active against some isolates of *P. aeruginosa* and certain *Proteus* spp. that are resistant to ampicillin and its congeners.
- Sensitive to destruction by β -lactamases and are generally less effective against Gm(+) infections than the other penicillins

➤ Two different sub-groups

- α -carboxypenicillins

- ✓ Carbenicillin, Ticarcillin

- Acylaminopenicillins

- ✓ Mezlocillin, Piperacillin

➤ Their use is generally limited to treatment of serious infections caused by susceptible **Gm(-)** bacilli or mixed aerobic-anaerobic infections

Mechanism of Resistance

- Decreased affinity PBPs
- Inability to penetrate to the site of action
 - ✓ Difference in susceptibility of Gm(-) Gm(+) bacteria
- Active efflux mechanisms (*P. aeruginosa*, *E. coli*, and *Neisseria gonorrhoeae*.)
- Enzymatic destruction of the β -lactam ring

Untoward effects

➤ Hypersensitivity reactions

- 0.7% and 4% of all treatment courses
- Maculopapular rash, urticarial rash, fever, bronchospasm, vasculitis, serum sickness, exfoliative dermatitis, Stevens-Johnson syndrome, and anaphylaxis

➤ Other adverse reactions

- bone marrow depression, granulocytopenia, and hepatitis
- Pain and sterile inflammatory reactions at the sites of intramuscular injections
- Nausea, vomiting, diarrhea
- Superinfection

Cephalosporins

Introduction

- Are derivatives of **7-aminocephalosporanic** acid
- Act in the same manner as penicillins
- Are relatively stable in dilute acid and are highly resistant to penicillinase

General Features of the Cephalosporins

- Many are absorbed readily after oral administration while some are administered parenterally
- Excretion primarily by the kidney (dosage adjustment in renal insufficiency)
 - ✓ Probenecid slows tubular secretion
- Several cephalosporins penetrate into **CSF** in sufficient concentration, cross the **placenta**, relatively good penetration into the **aqueous humor** but is poor into the **vitreous** humor

Drug	Route	Half-life (hr)	Renal Excretion (%)
First generation			
Cefadroxil	Oral	1.2–2.5	70–90
Cefazolin	IM, IV	1.5–2.5	70–95
Cephalexin	Oral	1.0	95
Cephapirin	IM, IV	0.6	50–70
Cephradine	Oral	0.7	75–100
Second generation			
Cefaclor	Oral	0.6–0.9	60–85
Cefamandole	IM, IV	0.5–1.2	100
Cefmetazole	IV	1.2–1.5	85
Cefonicid	IM, IV	3.5–4.5	95–99
Cefotetan	IM, IV	2.8–4.6	60–91
Cefoxitin	IM, IV	0.7–1.0	85
Cefprozil	Oral	1.2–1.4	64
Cefuroxime	IM, IV	1.1–1.3	95
Cefuroxime axetil	Oral	1.1–1.3	52
Loracarbef ^a	Oral	1.0	87–97
Third generation			
Cefdinir	Oral	1.7	18
Cefepime	IM, IV	2.0	70–99
Cefixime	Oral	2.3–3.7	50
Cefoperazone	IM, IV	2.0	20–30
Cefotaxime	IM, IV	1.0	40–60
Cefpodoxime proxetil	Oral	1.9–3.7	40
Ceftazidime	IM, IV	1.9	80–90
Ceftibuten	Oral	1.5–2.8	57–75
Ceftizoxime	IM, IV	1.4–1.8	57–100
Ceftriaxone	IM, IV	5.8–8.7	33–67

Classification

- Classified into four based on their spectrum of activity
 - First generation
 - ✓ Cefadroxil, cefazolin, cephalixin, Cephalothin
 - Second generation
 - ✓ Cefaclor, Cefprozil, Cefuroxime, Cefoxitin (a cephamycin)
 - Third generation
 - ✓ Cefdinir, Cefixime, Cefotaxime, Cefpodoxime, Ceftazidime, Ceftibuten, Ceftriaxone, Cefditoren
 - Fourth generation
 - ✓ Cefepime, Cefpirome

First Generation Cephalosporins

- Active in vitro against
 - Gm(+) cocci including Penicillinase-producing and nonpenicillinase producing Staphylococcal strains, streptococcus pyogenes, S. pneumonia
- Have limited activity against **Gm(-)** bacteria (some strains of E. coli, Klebsiella, Shigella are inhibited in vitro)
- Inactive against
 - *Enterococci*, **MRSA**, *Listeria monocytogenes*

Second generation cephalosporins

- Have greater stability against β -lactamase
- Active, in vitro, against
 - Bacteria susceptible to 1st GC
 - Most strains of *Haemophilus influenza* (anaerobes)
 - **Gm(-)** bacteria (2nd GC are more active than 1st GC)
 - May be active against *Enterobacter*, *E.coli*, *Klebsiella*, *Neisseria* that are resistant to 1st GC
- Inactive against
 - Enterococci, MRSA, and Pseudomonas

Third generation cephalosporins

- Higher β -lactamase stability
- Less active in vitro against susceptible staphylococci than 1st GC
- Expanded spectrum of activity against **Gm(-)**
- Active in vitro against
 - Gm(-) susceptible to 1st and 2nd G.C.s
 - Most are active against enterobacter, E. coli, Klebsiella, Neisseria.. That may be resistant to 1st and 2nd G.C.
 - Some are active against **Pseudomonas**
- Inactive against **MRSA** and generally are inactive against ***enterococci*** and ***Listeria monocytogenes***

Fourth generation cephalosporins

- More resistant to inactivation by β -lactamases
- Expanded spectrum of activity against **Gm(-)**
 - Active against some Gm(-) including *Pseudomonas aeruginosa* and certain *Enterobacteriaceae*, that are generally resistant to 3rd GC
- More active against **Gm(+)** than most 3rd GCs
- Inactive against MRSA, enterococci, and *Listeria monocytogenes*

Mechanisms of Bacterial Resistance to the Cephalosporins

- Resistance to the cephalosporins may be related to
 - Inability of the antibiotic to reach its sites of action
 - Alterations in the penicillin-binding proteins (PBPs)
 - **Destruction of the cephalosporins by hydrolysis of the β -lactam ring (by β -lactamases)**

Adverse reaction

➤ Hypersensitivity

- Anaphylaxis, bronchospasm, and urticaria
- Cross-reactivity with penicillin allergy (in 20%)

➤ Nephrotoxicity, Diarrhea, disulfiram-like reaction,

Therapeutic Uses

- Skin and soft tissue infections due to *S. aureus* and *S. pyogenes* (1st GCs)
- Colorectal surgery, where prophylaxis for intestinal anaerobes is desired (2nd GCs)
- Generally 2nd GCs displaced by 3rd GCs
 - **Oral 2nd GCs can be used to treat RTI**
 - In situations where facultative Gm(-) and anaerobes are involved (intra-abdominal infections, pelvic inflammatory disease, and diabetic foot infection)

Therapeutic Uses (cont'd)

- 3rd GCs (with or without **aminoglycosides**)
 - Drugs of choice for serious infections caused by *Klebsiella*, *Enterobacter*, *Proteus*, *Serratia*, and *Haemophilus* spp.
 - Ceftriaxone is the therapy of choice for all forms of gonorrhea and for severe forms of Lyme disease.
 - Drugs of choice for the treatment of meningitis caused by *H. influenzae*, sensitive *S. pneumoniae*, *N. meningitidis*, and gram-negative enteric bacteria
 - Treatment of ***Pseudomonas meningitis*** infection
 - Treatment of community-acquired pneumonia

Therapeutic Uses (cont'd)

➤ 4th GCs

- Empirical treatment of nosocomial infections where antibiotic resistance owing to extended-spectrum β -lactamases or chromosomally induced β -lactamases are anticipated

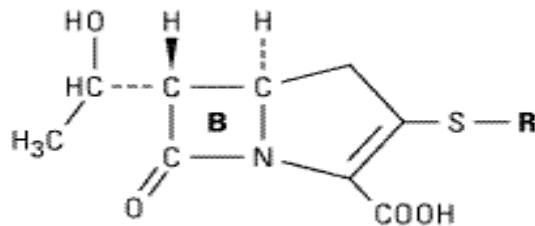
Other β -lactam antibiotics

Carbapenems

- Carbapenems are β -lactam antibiotics that contain a fused β -lactam ring and a five-membered ring system that differs from the penicillins in being unsaturated and containing a carbon atom instead of the sulfur atom.
- Has a broader spectrum of activity than do most other β -lactam antibiotics.

Imipenem

- Marketed in combination with ***cilastatin***, a drug that inhibits the degradation of imipenem by a renal tubular dipeptidase
- Derived from a compound produced by *Streptomyces cattleya*. The compound thienamycin is unstable, but imipenem is stable



Mechanism of Action

- Binds to penicillin-binding proteins, disrupts bacterial cell wall synthesis, and causes death of susceptible microorganisms.
- It is very resistant to hydrolysis by most β -lactamases.

Antimicrobial Activity

- Active against
 - *Streptococci* (including pen-resistant), *enterococci* (excluding *E. faecium* and non- β -lactamase-producing pen-resistant strains), *staphylococci* (including penicillinase-producing strains), and *Listeria*
- Most MRSA strains are resistant
- Excellent Activity against the *Enterobacteriaceae*, including organisms that are cephalosporin-resistant
- Most strains of *Pseudomonas* and *Acinetobacter* are inhibited.
- Anaerobes, including *B. fragilis*, are highly susceptible.

➤ **Pharmacokinetics**

- Not absorbed orally
- Hydrolyzed rapidly by dipeptidase in renal tubule
- Approximately 70% of administered imipenem is recovered in the urine as the active drug

➤ **Adverse Reactions**

- Nausea and vomiting (most common), Seizures, patients allergic to other β -lactam antibiotics may have hypersensitivity reactions

Therapeutic Uses

- Especially useful for the treatment of infections caused by cephalosporin-resistant nosocomial bacteria
- For empirical treatment of serious infections in hospitalized patients who have recently received other β -lactam
- Imipenem should not be used as monotherapy for infections caused by *P. aeruginosa* because of the risk of resistance developing during therapy

➤ Meropenem

- Not sensitive to renal dipeptidase.
- Similar toxicity to imipenem (less likely to cause seizures)
- It is therapeutically equivalence to imipenem.

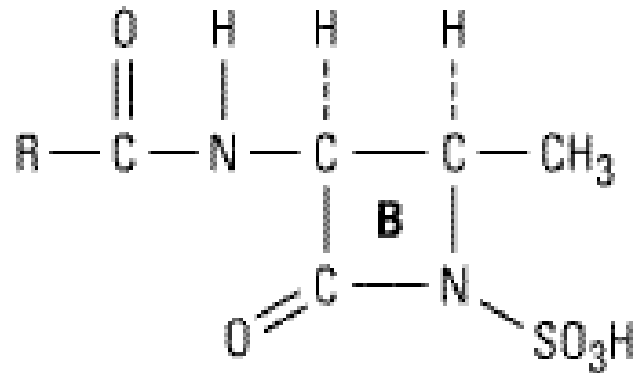
➤ Ertapenem

- Have larger serum $t_{1/2}$ that allows once-daily dosing
- Inferior activity against *P. aeruginosa* & *Acinetobacter* spp.

Monobactams

➤ Aztreonam

- A monocyclic β -lactam compound (a monobactam) isolated from *Chromobacterium violaceu*



Mechanism of Action and antimicrobial activity

- Acts in the same way as other β -lactam antibiotics
- Aztreonam is resistant to many of the β -lactamases elaborated by most Gm(-)
- Its antimicrobial activity differs from those of other β -lactams and resembles that of aminoglycosides.
- No activity against Gm(+) bacteria & anaerobic organisms
- Excellent activity against *Enterobacteriaceae* and *P. aeruginosa*.

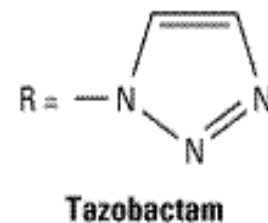
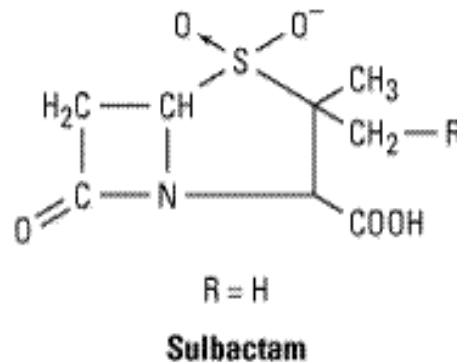
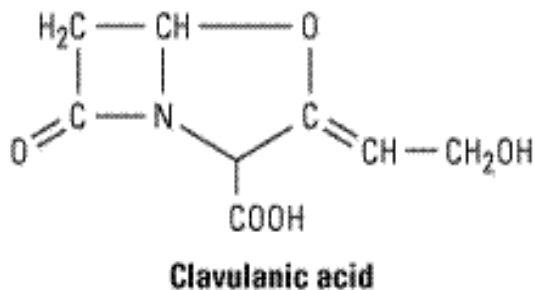
Therapeutic Uses

- Little allergic cross-reactivity with β -lactam antibiotics (possible exception of ceftazidime)
- Aztreonam is useful for treating g(-) infections that normally would be treated with a β -lactam antibiotic were it not for the history of a prior allergic reaction.

β -Lactamase Inhibitors

- β -Lactamase inhibitors are most active against plasmid-encoded β -lactamases, but are inactive at clinically achievable concentrations against β -lactamases induced in Gm(-) bacilli by treatment with 2nd and 3rd GCs
- They resemble β -lactam molecules but have very weak intrinsic antibacterial action.

- A "suicide" inhibitor that irreversibly binds β -lactamases produced by a wide range of gram-positive and gram-negative microorganisms
- Used in combination with β -lactam antibiotics to increase their spectrum of activity.
 - Amoxicillin + clavulanate (augmentin)
 - Ticarcillin + clavulanate (Timentin)



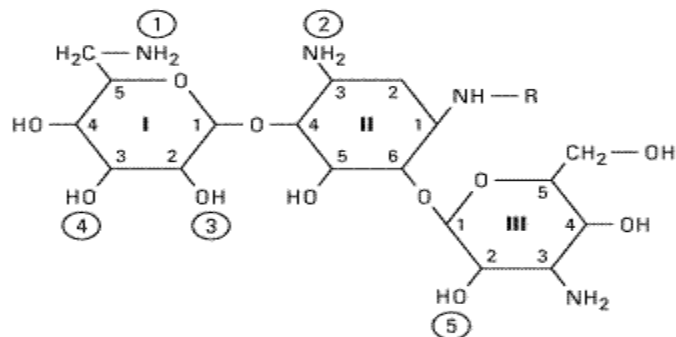
AMINOGLYCOSIDES

Introduction

- Gentamicin, Tobramycin, Amikacin, Netilmicin, Kanamycin, Streptomycin, Neomycin
- Primarily used to treat infections caused by aerobic gram-negative bacteria
- Streptomycin: in the treatment of tuberculosis
- Have bactericidal activity

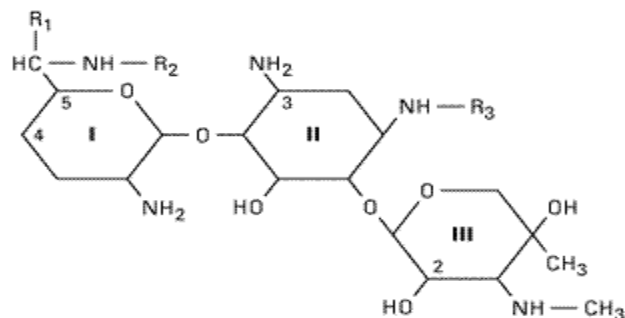
Chemistry

- The aminoglycosides consist of
 - A hexose (aminocyclitol) nucleus [either streptidine (found in streptomycin) or 2-deoxystreptamine (found in all other available aminoglycosides)]
 - Amino sugars attached through glycosidic linkages to the aminocyclitols
- Are water-soluble, stable in solution, and more active at alkaline than at acid pH



Kanamycin R = H

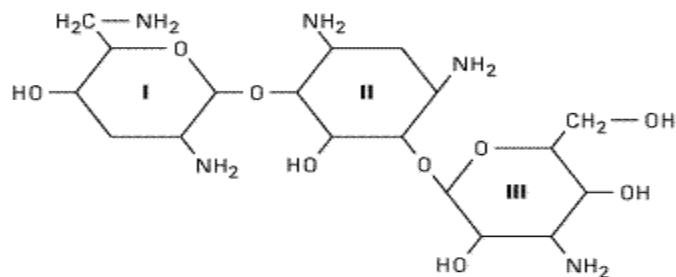
Amikacin R = $\text{C}(=\text{O})-\text{CH}(\text{OH})-\text{CH}_2-\text{CH}_2-\text{NH}_2$



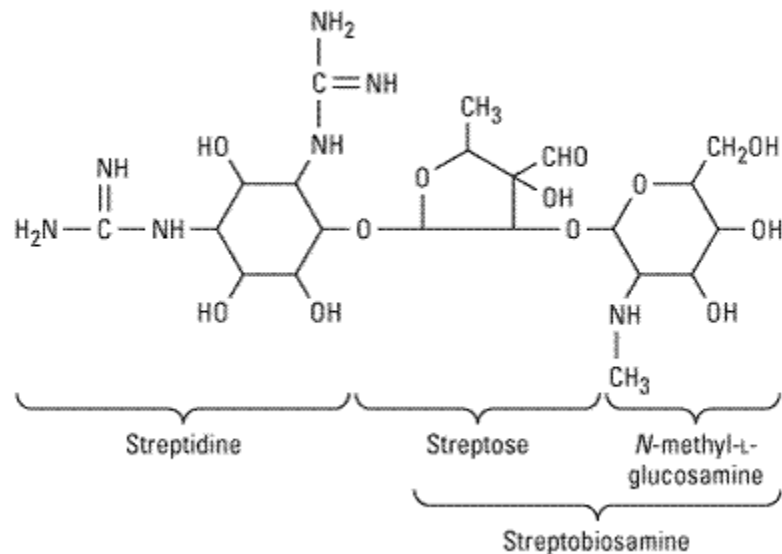
Gentamicin, netilmicin

Gentamicin C₁
Gentamicin C₂
Gentamicin C_{1a}
Netilmicin

Ring I			Ring II
R ₁	R ₂	C4-C5 bond	R ₃
CH ₃	CH ₃	Single	H
CH ₃	H	Single	H
H	H	Single	H
H	H	Double	C ₂ H ₅



Tobramycin



Streptomycin

Mechanism of action

➤ Aminoglycosides

- Are rapidly bactericidal
- Have a **post-antibiotic** effect (their killing action continues when their plasma levels have declined below measurable levels)

➤ Penetration through the outer bacterial membrane

- Outer membrane disruption
- Diffusion through outer membrane porins.

Mechanism of action (cont'd)

- Transport across the **cytoplasmic** membrane depends on **electron transport**
 - Requirement for a membrane electrical potential (interior negative) to drive permeation
 - Is rate-limiting and can be blocked or inhibited by divalent cations (*e.g.*, Ca^{2+} and Mg^{2+}), hyperosmolarity, a **reduction** in **pH**, and **anaerobic** conditions.
 - Antimicrobial activity reduced in the **anaerobic** environment of an abscess and in hyperosmolar acidic urine

Mechanism of action (cont'd)

➤ Inside the cell

- Aminoglycosides bind to ribosomes and interfere with protein synthesis
 - ✓ misreading and premature termination of mRNA translation
- Hence, aberrant proteins produced and may be inserted into the cell membrane, leading to altered permeability

➤ Bind to 30S ribosomal subunit and disrupt the normal cycle of ribosomal function by

- blocking formation of **initiation complex**
- **Misreading** of the mRNA template
- Incorporating incorrect AA to growing polypeptide chains
- inhibiting translocation

Mechanism of Resistance

- Resistance could be due to
 - Failure to penetrate intracellularly
 - Low affinity of the drug for the bacterial ribosome
 - Inactivation of the drug by microbial enzymes
 - ✓ Aminoglycoside-modifying enzymes **phosphorylate, adenylate,** or **acetylate** specific hydroxyl or amino groups
 - ✓ The metabolites may compete with the parent drug for transport across the inner membrane, but are incapable of binding to ribosomes

Antibacterial Spectrum

- Primarily active against aerobic **Gm(-)** bacilli
- Have little activity against anaerobic microorganisms or facultative bacteria under anaerobic conditions.
- Limited action against most gram-positive bacteria
 - Produce synergistic bactericidal effect *in vitro* (against enterococci, streptococci, and staphylococci) when combined with a cell wall-active agent

Pharmacokinetics

➤ Absorption

- Very poorly absorbed from the GIT.
- Absorbed rapidly from IM sites of injection.
 - ✓ In critically ill patients, especially those in shock, absorption of drug may be reduced from intramuscular sites because of poor perfusion

➤ Distribution

- Do not penetrate into most cells and has negligible plasma albumin binding (Except for streptomycin)
- $V_d \cong$ the volume of extracellular fluid
- High concentrations are found in renal cortex and inner ear (likely contributor to the nephrotoxicity and ototoxicity)
- **Streptomycin** and **tobramycin** can cause hearing loss in children born to women who receive the drug during pregnancy.
 - ✓ Recommendation: use with caution during pregnancy and only for strong clinical indications in the absence of suitable alternatives.

➤ Elimination.

- The aminoglycosides are excreted almost entirely by glomerular filtration,

➤ Dosing

- Total daily dose as a single injection or two or three equally divided doses
- Total daily dose as a single injection is associated with less toxicity and is just as effective as multiple-dose regimens
 - ✓ More drug accumulates with higher plasma concentrations, particularly at trough, and with prolonged periods of exposure.

Untoward effects

➤ **Vestibular & auditory ototoxicity**

- Vestibular and auditory dysfunction can follow the administration of any of the aminoglycosides

➤ **Nephrotoxicity**

➤ **Neuromuscular Blockade**

- And the associated apnea
- Decreasing potency for blockade: neomycin, kanamycin, amikacin, gentamicin, and tobramycin.

Streptomycin

- Is generally less active than other members of the class
- Administered by deep IM or IV
- Dose adjustment in renal impairment
- Therapeutic Uses
 - ***Bacterial Endocarditis***
 - ✓ Streptomycin and penicillin in combination
 - ***Tularemia***
 - ✓ Streptomycin (or gentamicin) is the drug of choice
 - ***Plague***
 - ***Tuberculosis***

Gentamicin

- First choice due to low cost and reliable activity against all but the most resistant g(-) aerobes.
- Available for parenteral, ophthalmic, and topical
- Therapeutic Uses (Gentamicin, tobramycin, amikacin, and netilmicin)
 - *Urinary Tract Infections*
 - *Pneumonia*
 - *Meningitis*
 - *Bacterial Endocarditis*
 - *Sepsis*

Tobramycin

➤ Therapeutic Uses

- Same as those for gentamicin
- The preferred aminoglycoside for treatment of serious infections caused by *P. **aeuroginosa*** (usually in combination with an antipseudomonal β -lactam antibiotic)
- Poor activity in combination with penicillin against many strains of enterococci.
- Ineffective against mycobacteria

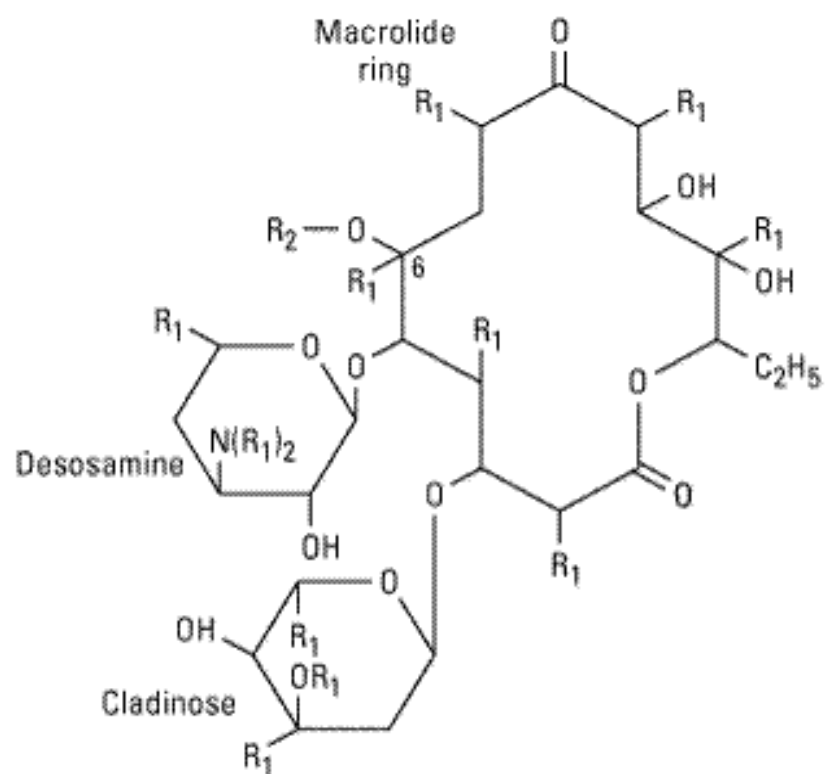
Amikacin

- Broadest spectrum of activity in the group
- Resistant to many of the aminoglycoside-inactivating enzymes
- Therapeutic Uses
 - The preferred agent for the initial treatment of serious nosocomial Gm(-) bacillary infections in hospitals
 - Active against the majority of aerobic g(-) bacilli in the community and the hospital
 - Less active than gentamicin against enterococci
 - It is active against *M. Tuberculosis*

MACROLIDES

Introduction

- *Contain a large 14 or 15 membered lactone ring to which one or more deoxy sugars are attached.*
- Include erythromycin, *clarithromycin, azithromycin, and oleandomycin*
- Clarithromycin differs from erythromycin only by methylation of the hydroxyl group at the 6 position,
- Azithromycin differs by the addition of a methyl-substituted nitrogen atom into the lactone ring.
 - Improve acid stability and tissue penetration and broaden the spectrum of activity



Erythromycin ($R_1 = CH_3$, $R_2 = H$)

Clarithromycin ($R_1, R_2 = CH_3$)

Antibacterial Activity

- Erythromycin usually is **bacteriostatic**, but may be **bactericidal** in high concentrations
- The antibiotic is most active *in vitro* against aerobic gram-positive cocci and bacilli.
- Macrolide resistance is common among streptococci.
- **Cross-resistance** among macrolides is complete
- Staphylococci are not reliably sensitive to erythromycin.

Antibacterial Activity (cont'd)

- Erythromycin is **inactive** against most aerobic enteric Gm-negative bacilli.
- Macrolides are active against *Campylobacter jejuni*.
- Erythromycin is active against *M. Pneumoniae* and *Legionella pneumophila*

Antibacterial Activity (cont'd)

- Clarithromycin is more potent than erythromycin against sensitive strains of streptococci and staphylococci, and has modest activity against *H. influenzae* and *N. gonorrhoeae*.
- Azithromycin: less active than erythromycin against Gm(-) organisms and slightly more active against *H. influenzae* and *Campylobacter* spp.
- Azithromycin & clarithromycin have enhanced activity against *M. avium-intracellulare* & some protozoa (*e.g.*, *T. gondii*, *Cryptosporidium*, and *Plasmodium* spp.).

Mechanism of Action

- Inhibit protein synthesis by binding reversibly to **50S** ribosomal subunit
- Inhibits the **translocation** step where in a newly synthesized peptidyl tRNA molecule moves from the **acceptor** site on the ribosome to the **peptidyl donor site**.
- **Increased** antimicrobial activity at alkaline pH (more permeability of the un-ionized form)

Mechanism of resistance

- Resistance to macrolides usually results from
 1. Drug efflux by an active pump mechanism
 2. Ribosomal protection by inducible or constitutive production of **methylase** enzymes, which modify the ribosomal target and decrease drug binding
 3. Macrolide hydrolysis by **esterases** produced by Enterobacteriaceae
 4. Chromosomal mutations that alter a 50S ribosomal protein

Pharmacokinetics

- Absorbed from the intestinal tract,
- Presence of food interferes with absorption (Azithromycine)
- Erythromycin is relatively more acid labile
- Serum levels of clarithromycin and azithromycin are low, but they concentrate in tissue and reach high levels.
- Erythromycin and azithromycin are excreted primarily in bile
- **Clarithromycin** is metabolized to the biologically active 14-OH metabolite and is eliminated largely by the kidney.
- $t_{1/2}$: erythromycin (1.4 hours), clarithromycin (3 to 7), azithromycin (68 hours).

Clinical uses

- *Mycoplasma pneumonia* infections
 - A macrolide or TTC is the drug of choice
- *Legionnaires' Disease*
 - Azithromycin or a fluoroquinolone
- *Chlamydial* Infections
 - Can be treated effectively with any of the macrolides
- Diphtheria
 - erythromycin
- Pertussis
 - Drug of choice for treatment & prophylaxis

Clinical uses (cont'd)

- Streptococcal infections
 - Alternatives in patients who are allergic to penicillin. Penicillin-resistant strains of *S. Pneumoniae* are very likely to be resistant to macrolides
- Staphylococcal infections
- Campylobacter infections
 - largely replaced by Fluoroquinolones in adults
 - Erythromycin remains useful for treatment of *Campylobacter* gastroenteritis in children.

Clinical uses (cont'd)

➤ ***Tetanus***

- Erythromycin may be given to eradicate ***Clostridium tetani*** in patients allergic to penicillin

➤ ***Mycobacterial Infections***

- Clarithromycin or azithromycin is recommended as first-line therapy for prophylaxis and treatment of disseminated infection caused by ***M. avium-intracellulare*** in AIDS patients and for treatment of pulmonary disease in non-HIV-infected patients

Erythromycin, 1st choice	Alternative
Mycoplasma pneumoniae	Fluoroquinolone
Legionella pneumoniae	Doxycycline + Rifampin or cotrimoxazole
Diphtheria	Penicilline G
Pertussis	Cotrimoxazole
Chlamydia trachomatis [pneumonia; conjunctivitis]	Sulfisoxazole
Campylobacter jejuni [Gastroenteritis]	Tetracycline
Bacillary angiomatosis	Doxycycline

Infections in which erythromycin is used as alternative drug	Drug of 1st choice
Gps A,C,G streptococcal infection	Penicilline G
Streptococcus pneumoniae infection	Penicillin G, Ceftriaxone
Rheumatic fever prophylaxis	Penicilline G
Anthrax	Penicilline G
Lymphogranuloma venereum	Tetracycline
Chlamydia trachomatis, Urethritis, Cervicitis	Azithromycin, Tetracycline
Acne vulgaris	Tetracycline p.o.; other topical agents

Adverse Effects

- The incidence of side effects associated with erythromycin therapy is very low.
 - Mild GI upset with nausea, diarrhea, and abdominal pain (more common)
 - Rashes are seen infrequently
 - Thrombophlebitis may follow IV administration
 - Transient impairment of hearing.
- **Cholestatic hepatitis** (characterized by fever, enlarged and tender liver, hyperbilirubinemia, dark urine, eosinophilia, elevated serum bilirubin, and elevated transaminase levels)
- The drug inhibits hepatic microsomal enzymes and interfere with the actions of various drugs

KETOLIDES (Telithromycin)

- are semisynthetic derivatives of erythromycin, differing from erythromycin by substitution of a 3-keto group for the neutral sugar L-cladinose
- This renders ketolides less susceptible to methylase-mediated and efflux-mediated mechanisms of resistance.
- Ketolides therefore are active against many macrolide-resistant g(+) strains.

➤ Antibacterial Activity

- Have similar antibacterial properties with macrolides
- Telithromycin is active against *staphylococci*, *streptococci*, *S. pneumoniae*, *Haemophilus* spp., *Moraxella catarrhalis*, *mycoplasma*, *chlamydia*, and *Legionella*
- It is slightly more active by weight than erythromycin.

Pharmacokinetics

- Oral formulated (well absorbed, 60% bioavailability)
- $t_{1/2}$ 9.8 hours (once daily administration)
- 60% to 70% bound by serum protein
- Penetrates well into most tissues
- Telithromycin is concentrated into macrophages and white blood cells
- Primarily cleared by hepatic metabolism

➤ **Therapeutic Uses**

- Approved for treatment of RTI, including acute exacerbation of chronic bronchitis, acute bacterial sinusitis (5-day regimen), and community-acquired pneumonia.

➤ **Untoward Effects**

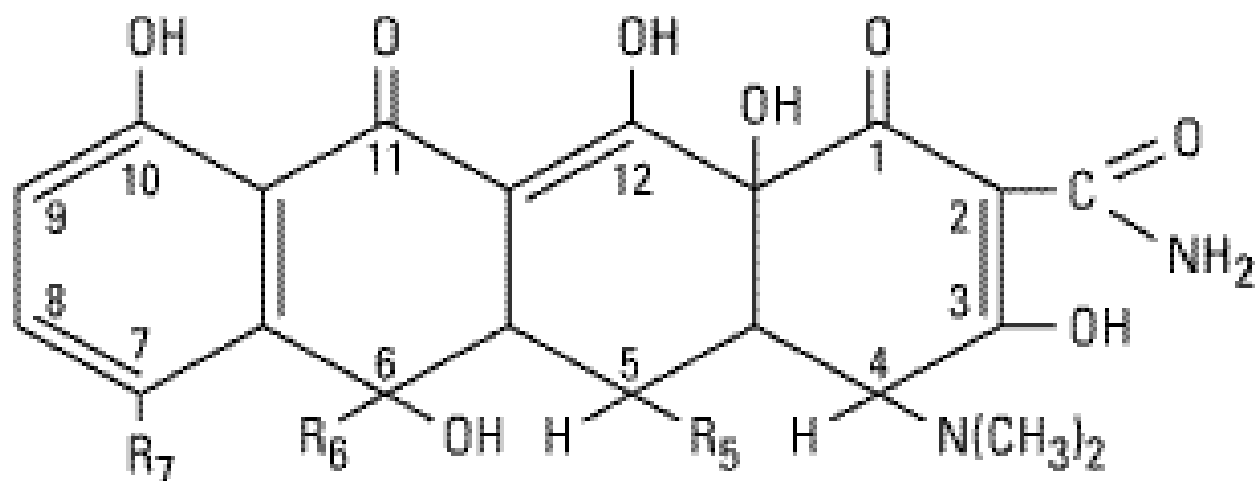
- Telithromycin generally is well tolerated
- Nausea, vomiting, and diarrhea (most common side effects)
- Visual disturbances, reversible hepatic dysfunction, pseudomembranous colitis, reports of disease exacerbation in patients with myasthenia gravis
- Clinically significant QTc prolongation
- Telithromycin has several clinically significant drug interactions similar to those for erythromycin. It is both a substrate and a strong inhibitor of CYP3A4.

TETRACYCLINES

(TTC)

Introduction

- Produced by different species of *Streptomyces*
 - *OxyTTC* is a natural product elaborated by *Streptomyces rimosus*
 - TTC is a semisynthetic derivative of chlorTTC
 - *Demeclocycline* is the product of a mutant strain of *Strep. Aureofaciens*
 - *Methacycline*, *doxycycline*, and *minocycline* are semisynthetic derivatives.
- *Chlortetracycline*, the prototype of the class, was introduced in 1948



				Renal Clearance (mL/min)
	R₇	R₆	R₅	
Chlortetracycline	— Cl	— CH ₃	— H	35
Oxytetracycline	— H	— CH ₃	— OH	90
Tetracycline	— H	— CH ₃	— H	65
Demeclocycline	— Cl	— H	— H	35
Methacycline	— H	= CH ₂ *	— OH	31
Doxycycline	— H	— CH ₃ *	— OH	16
Minocycline	— N(CH ₃) ₂	— H	— H	10

*There is no — OH at position 6 on methacycline and doxycycline.

Effects on Pathogenic Microorganisms

- Are bacteriostatic antibiotics active against a wide range of **aerobic** and **anaerobic** Gm(+) & Gm(-) bacteria
- Are also effective against ***Rickettsia***, ***Coxiella burnetii***, ***Mycoplasma pneumoniae***, ***Chlamydia*** spp., ***Legionella*** spp., ***Ureaplasma***, some atypical mycobacteria, and *Plasmodium* spp., that are resistant to cell-wall-active antimicrobial agents

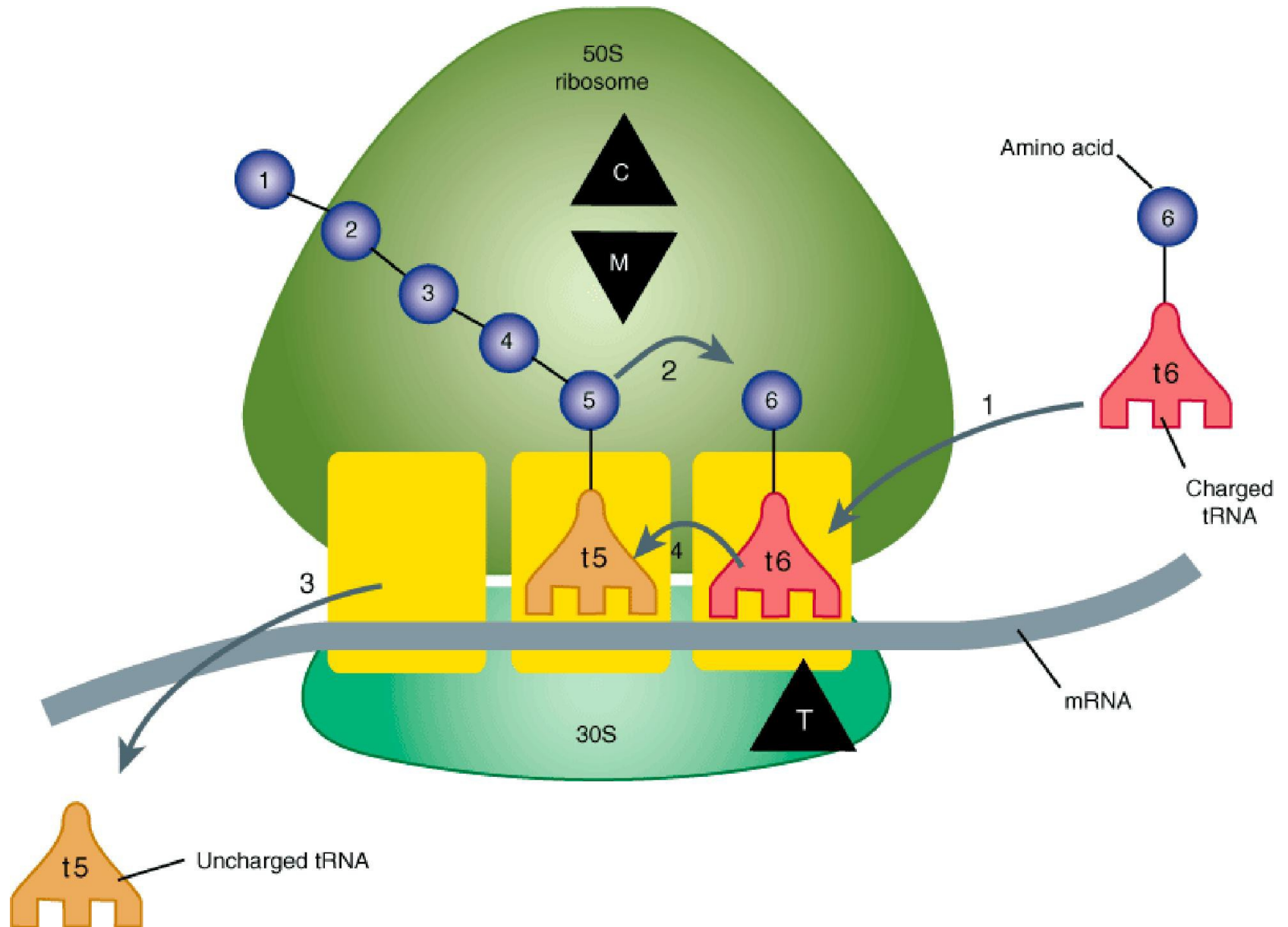
Effects on Intestinal Flora

- Many of the TTCs are incompletely absorbed from the GIT, and high concentrations in the bowel can alter enteric flora.
- Sensitive aerobic & anaerobic coliform microorganisms & Gm(+) spore-forming bacteria are markedly suppressed
- Overgrowth of TTC-resistant microorganisms (yeasts (*Candida* spp.), enterococci, *Proteus*, & *Pseudomonas*)

Mechanism of Action

- TTCs inhibit bacterial protein synthesis by binding to the **30S** bacterial ribosome and preventing access of aminoacyl tRNA to the acceptor (A) site on the mRNA-ribosome complex
- TTCs enter Gm(-) bacteria by passive diffusion through the porin channels of the outer cell membrane and by active transport *via* an energy-dependent system that pumps all TTCs across the cytoplasmic membrane
- Entry into Gm(+) bacteria requires metabolic energy, but is not as well understood

M.O.A. TTC



Resistance to the Tetracyclines

➤ The main resistance mechanisms

1. Decreased accumulation of tetracycline as a result of either decreased antibiotic influx or acquisition of an energy-dependent efflux pathway;
2. Production of a ribosomal protection protein that displaces tetracycline from its target
3. Enzymatic inactivation of tetracyclines

Pharmacokinetics

➤ Absorption

- Partially absorbed from the stomach and upper GIT
- Impaired by Food (*all except doxycycline and minocycline*)
- *TTCs form insoluble chelates with calcium, magnesium, and other metal ions* ⇒ interfere with absorption

➤ Distribution

- Distributed throughout body tissues and fluids
 - ✓ Minocycline and doxycycline (highest lipid solubility), oxyTTC (the least)

➤ Elimination

- TTCs are excreted mainly in bile and urine
- Ten to 50 % of various TTCs is excreted into the urine.
- Doxycycline is eliminated by nonrenal mechanisms ⇒ TTC of choice in renal insufficiency.

Classification

- Based on serum half-lives, TTCs are Classified
 - Short-acting
 - ✓ Serum half-life of 6-8 hours
 - ✓ ChlorTTC, TTC, oxyTTC
 - Intermediate-acting)
 - ✓ Serum half-life of 12 hours
 - ✓ Demeclocycline and methacycline
 - long-acting
 - ✓ Serum half-life of 16-18 hours
 - ✓ Doxycycline and minocycline
 - ✓ Almost complete absorption and slow excretion of these drugs allow for once daily dosing.

Therapeutic Uses

➤ ***Rickettsial Infections***

- Doxycycline is the drug of choice

➤ **Mycoplasma Infections**

➤ ***Chlamydial infections***

➤ ***Trachoma***

➤ ***Sexually Transmitted Diseases***

- Because of resistance, doxycycline no longer is recommended for gonococcal infections

➤ ***Anthrax***

- Doxycycline: for prevention or treatment of anthrax

Therapeutic Uses (cont'd)

➤ ***Bacillary Infections***

- Brucellosis
 - ✓ Tetracyclines in combination with rifampin or streptomycin
- Tularemia
 - ✓ TTCs are effective, though streptomycin is preferable
- Cholera
 - ✓ Doxycycline

➤ ***Other Bacillary Infections***

- Ineffective in infections caused by *Shigella*, *Salmonella*, or other Enterobacteriaceae

Therapeutic Uses (cont'd)

➤ ***Coccal Infections***

- Resistance restricts their use in staphylococcal, streptococcal, or meningococcal infections.
- Community strains of MRSA often are susceptible to TTC, doxycycline, or minocycline
- Doxycycline effective for empirical therapy of community-acquired pneumonia

➤ Are generally drugs of choice in infections with *Mycoplasma pneumoniae*, *chlamydiae*, *rickettsiae*, and some spirochetes

➤ Are sometimes employed in the treatment of protozoal infections (*Entamoeba histolytica*, *Plasmodium falciparum*)

Major Indications	Effective alternative
1. Chlamydial Infections Lymphogranuloma venereum Conjunctivitis Trachoma Acute epididymitis 2. Ricketisial infections Typhus fever Rocky mountain spotted fever 3. Other infections Plague Tularemia Brucellosis Relapsing fever Cholera Urethritis In combination with Aminoglycosides	- Acne, severe - <i>M. pneumoniae</i> - <i>Nocardia</i> - Rat-bite fever - Syphilis - Amebiasis - <i>P. falciparum</i> - Meningococcal disease - Prevention [minocycline] - Oral bowel preparations for Intestinal surgery [TTC+ Neomycin] - tx of inappropriate secretion Of ADH [Demeclocycline]

Adverse effects

➤ Gastrointestinal Adverse Effects

- Nausea, vomiting, and diarrhea
- modification the normal flora with overgrowth of pseudomonas, proteus, staphylococci, resistant coliforms, clostridia, and candida

➤ Bony Structures and Teeth

- Deposition in calcium deposited in newly formed bone or teeth

Adverse effects (cont'd)

➤ Liver Toxicity

- Especially during pregnancy

➤ Kidney Toxicity

- Aggravate azotemia in patients with renal disease

➤ Photosensitization

➤ Vestibular Reactions

- Dizziness, vertigo, nausea, and vomiting

➤ **Contraindications**

- In pregnant women
- Children under the age of 8 years

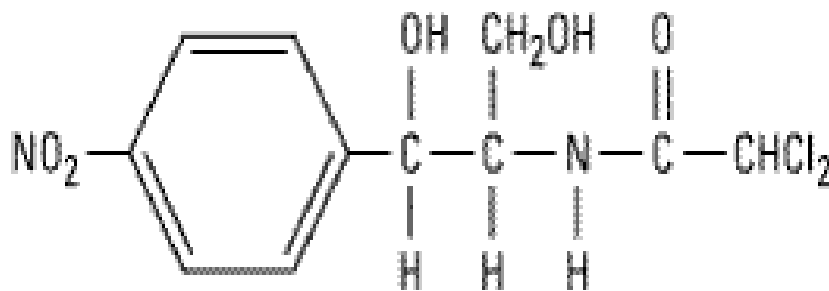
➤ Therapeutic selection

- Little difference in clinical response among the various TTCs.
- *The selection of an agent, therefore, is based on*
 - ❖ Tolerance
 - ❖ Ease of administration, and
 - ❖ Cost

CHLORAMPHENICOL

Introduction

- Crystalline chloramphenicol is a neutral, stable compound

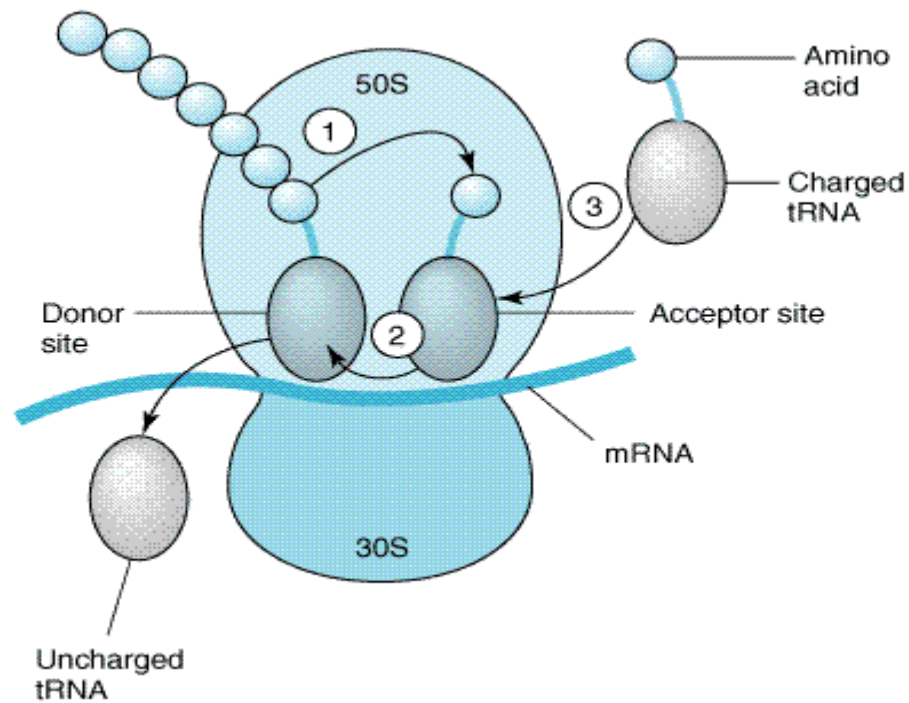


Chloramphenicol

- Contains a **nitrobenzene** moiety and is a derivative of **dichloroacetic acid**.
- The biologically active form is **levorotatory**.

Mechanism of Action

- It inhibits protein synthesis in bacteria, and to a lesser extent, in eukaryotic cells.
 - Binds reversibly to the 50S ribosomal subunit
 - Although binding of tRNA at the codon recognition site on the 30S ribosomal subunit is undisturbed, the drug **apparently prevents the binding of the amino acid-containing end of the aminoacyl tRNA to the acceptor site on the 50S ribosomal subunit.**
 - The interaction between **peptidyltransferase** and its amino acid substrate cannot occur, and **peptide bond formation is inhibited**



- Steps in bacterial protein synthesis and targets of (1) chloramphenicol; (2) macrolides, clindamycin, and type B streptogramins; and (3) tetracyclines

Pharmacokinetics

- Crystalline CAF is rapidly and completely absorbed
After oral administration
- widely distributed to virtually all tissues and body fluids, including the CNS and CSF s
- It penetrates cell membranes readily
- Most of the drug is inactivated either by conjugation with glucuronic acid or by reduction to inactive aryl amines

Antimicrobial Actions

- Broad spectrum antimicrobial activity
- Effective against **Gm(+)** and **Gm(-)** bacteria, including *Rickettsia*, *Mycoplasma*, and *Chlamydia* spp.
- Also effective against most **anaerobic** bacteria, including *Bacteroides fragilis*.

Resistance to Chloramphenicol

- Resistance is caused due to
 - Plasmid-encoded **acetyltransferase** that inactivates the drug (Acetylated derivatives fail to bind to bacterial ribosomes)
 - **Decreased permeability of the drug**
 - **Ribosomal mutation.**

Therapeutic Uses

➤ Typhoid Fever

- ✓ *3rd GC and quinolones are drugs of choice*

➤ Bacterial meningitis

- ✓ 3rd GC have replaced CAF
- ✓ An alternative in patients allergic to β -lactams and in developing countries

➤ Rickettsial Diseases

- ✓ The TTCs are the preferred agents
- ✓ In patients **allergic** to TTCs, in **reduced renal function**, in **pregnant women**, and in **children younger than 8** years of age who require prolonged or repeated courses of therapy, CAF is the drug of choice

➤ Brucellosis

- When a TTC is contraindicated CAF is recommended.

Untoward effects

- CAF inhibits the synthesis of proteins of the inner mitochondrial membrane, probably by inhibiting the ribosomal peptidyltransferase.
- These include subunits of cytochrome *c* oxidase, ubiquinone-cytochrome *c* reductase, and the proton-translocating ATPase critical for aerobic metabolism.
- Much of the toxicity observed with this drug can be attributed to these effects.

Untoward effects (cont'd)

- Newborn infants cannot adequately **conjugate** CAF to form the **glucuronide**
 - ⇒ High levels of free CAF cause the *gray baby syndrome*.
 - Abdominal distention, vomiting, progressive cyanosis, irregular respiration, hypothermia, and vasomotor collapse
- Effects on hematopoietic system: manifested by **bone marrow depression** or idiosyncratic aplastic anemia.
 - Bone marrow depression is dose related and is characterized by **anemia**, **leukopenia** or **thrombocytopenia**, but it is reversible on discontinuation of CAF.
 - Aplastic anemia is usually fatal. The mechanism is not known, **but it occurs most frequently with oral or ocular administration.**

Drug Interactions

- CAF inhibits hepatic CYP450s and prolongs the $t_{1/2}$ of drugs metabolized by this system (*warfarin, dicumarol, phenytoin, chlorpropamide, antiretroviral protease inhibitors, rifabutin, and tolbutamide*)
- *Phenobarbital, rifampin* potently induce CYPs and shorten the $t_{1/2}$ of CAF

MISCELLANEOUS ANTIBIOTICS

Vancomycin

Introduction

- A glycopeptide antibiotic of molecular weight 1500 daltons produced by *Streptococcus orientalis*.
- With the single exception of flavobacterium, it is active only against gram-positive bacteria, particularly *staphylococci*.
- It is water-soluble and quite stable.

Antibacterial Activity

- a narrow-spectrum agent active against Gm(+) organisms
- **Bacteriostatic** against staphylococci, streptococci, and enterococci.
- G(+) rods, such as *Bacillus anthracis*, *Corynebacterium diphtheriae*, *Clostridium tetani*, and *Clostridium perfringens*, are also susceptible
- Not effective against Gm(-) rods, mycobacteria, or fungi.

Mechanism of Action

➤ Cell wall synthesis inhibitor

- Binds firmly to the D-Ala-D-Ala terminus of nascent peptidoglycan pentapeptide . This inhibits the **transglycosylase**, preventing further elongation of peptidoglycan and cross-linking.
- The peptidoglycan is thus weakened and the cell becomes susceptible to lysis.
- The cell membrane is also damaged, which contributes to the antibacterial effect.

Mechanism of Resistance

- Resistance to vancomycin in enterococci and vancomycin-resistant *S aureus strains* is due to
 - modification of the D-Ala-D-Ala binding site of the peptidoglycan building block in which the terminal D-Ala is replaced by D-lactate.

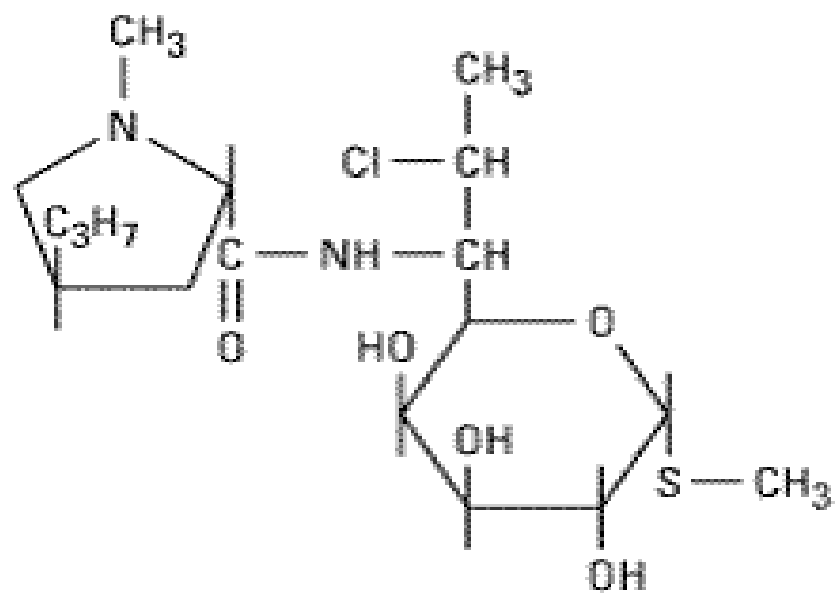
Pharmacokinetics

- Poorly absorbed from the GIT and should be orally administered only for the treatment of antibiotic-associated enterocolitis caused by *Clostridium difficile*
- Parenteral doses must be administered IV
- The drug has a serum elimination half-life of about 6 hours.
- Appears in various body fluids, including the CSF when the meninges are inflamed
- About 90% is excreted by glomerular filtration (dosage adjustments in renal impairment)

Clinical Uses

- Should be employed only to treat serious infections
- is particularly useful in the management of infections due to **MRSA**, and in severe staphylococcal infections in patients who are allergic to penicillins and cephalosporins.
- Effective and convenient when there is disseminated **staphylococcal infection or localized infection** of a shunt in a patient with irreversible renal disease who is being maintained by **hemodialysis** or **peritoneal dialysis**
- Treatment of infections caused by *Corynebacterium* spp.
- Important in the management of known or suspected penicillin-resistant pneumococcal infections

Clindamycin



Clindamycin

Mechanism of Action and Resistance

- Clindamycin inhibits protein synthesis by interfering with the formation of **initiation complexes** and with **aminoacyl translocation reactions**.
- The binding site for clindamycin on the **50S** subunit of the bacterial ribosome is identical with that for erythromycin.
- Resistance to clindamycin, which generally confers cross-resistance to other macrolides, is due to
 - i. mutation of the ribosomal receptor site
 - ii. modification of the receptor by a constitutively expressed methylase
 - iii. Enzymatic inactivation of clindamycin
- **Gm(-)** aerobic species are intrinsically resistant because of **poor permeability** of the outer membrane.

Mechanism of Action and Resistance (cont'd)

- ☞ Macrolide resistance due to ribosomal methylation may produce resistance to clindamycin.
- ☞ However, because clindamycin does not induce the methylase, there is cross-resistance only if the enzyme is produced constitutively.
- ☞ Clindamycin is not a substrate for macrolide efflux pumps

Antibacterial Activity

- Streptococci, staphylococci, and pneumococci are inhibited by clindamycin
- Enterococci and Gm(-) aerobic organisms are resistant (in contrast to their susceptibility to erythromycin).
- Bacteroides species and other anaerobes, both Gm(+) and Gm(-), are usually susceptible.
- *Clostridium difficile* is resistant.

Pharmacokinetics

➤ Absorption

- Nearly completely absorbed following oral administration
- Food in the stomach does not reduce absorption significantly

➤ Distribution

- Widely distributed in many fluids and tissues
- Readily crosses the placental barrier
- > 90% is bound to plasma proteins

➤ Excretion

- 10% is excreted unaltered in the urine, and small quantities are found in the feces.
- Inactivated by metabolism to **N-demethylclindamycin** and **clindamycin sulfoxide**, which are excreted in the urine and bile

Therapeutic Uses

- **Treatment of severe anaerobic infection** caused by bacteroides and other anaerobes that often participate in mixed infections.
- In combination with aminoglycoside or cephalosporin,
 - To treat **penetrating wounds** of the abdomen and the gut
- Clindamycin plus primaquine is an effective alternative to co-trimoxazole for moderate to moderately severe *Pneumocystis carinii pneumonia* in AIDS patients.
- Also used in combination with pyrimethamine for AIDS-related toxoplasmosis of the brain.

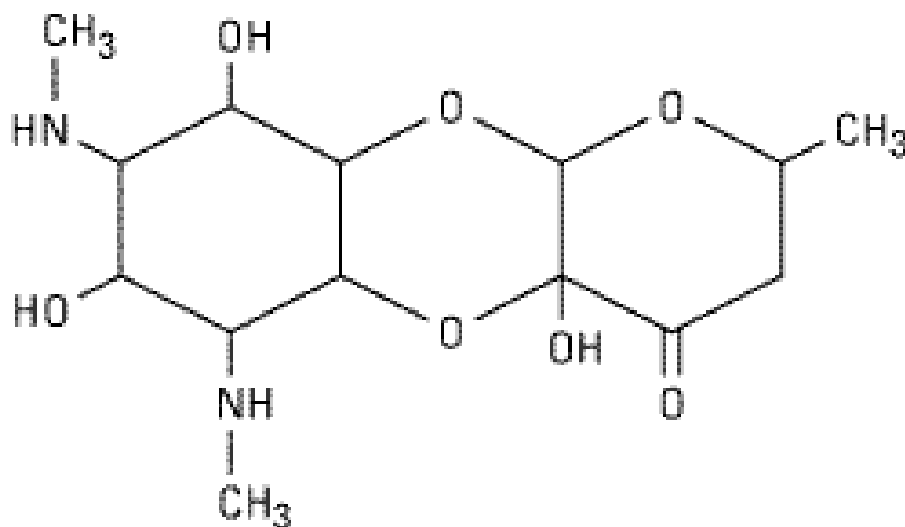
Untoward Effects

- **Diarrhea** (in 2% to 20% subjects)
- **Pseudomembranous colitis** caused by the toxin from *C. difficile*
 - Characterized by abdominal pain, diarrhea, fever, and mucus and blood in the stools.
 - *It may be lethal*
 - Discontinuation of the drug, combined with administration of metronidazole orally or intravenously usually is curative
- Skin rashes, Stevens-Johnson syndrome, impaired liver function, granulocytopenia, thrombocytopenia, and anaphylactic reactions
- Local thrombophlebitis with IV administration
- Can inhibit neuromuscular transmission and may potentiate the effect of a **neuromuscular blocking**

Spectinomycin

Introduction

- Spectinomycin is an antibiotic produced by *Streptomyces spectabilis*.
- It is an aminocyclitol with the following structural formula



Spectinomycin

Antibacterial Activity and Mechanism

- Active against a number of Gm(-) bacterial species, but it is inferior to other drugs to which such microorganisms are susceptible.
- *Its only therapeutic use is in the treatment of gonorrhea caused by strains resistant to first-line drugs, or if there are contraindications to the use of these drugs*

Mechanism of Action and Resistance

- Spectinomycin selectively inhibits protein synthesis in **Gm(-)** bacteria.
- It binds to and acts on the **30S** ribosomal subunit.
- Its action is similar to that of the **aminoglycosides**, but **spectinomycin** is not bactericidal and does not cause misreading of mRNA.
- Bacterial resistance may be mediated by mutations in the 16S ribosomal RNA or by modification of the drug by adenylyltransferase

➤ Pharmacokinetics

- Rapidly absorbed after IM injection.
- It is not significantly bound to plasma protein
- All of an administered dose is recovered in the urine within 48 hours.

➤ Therapeutic Uses

- Recommended in the treatment **uncomplicated gonococcal infection as an alternative** regimen in patients who are intolerant or allergic **to β -lactam antibiotics and quinolones.**

Untoward Effects

- Urticaria, chills, and fever, dizziness, nausea, and insomnia.
- Rarely Nephrotoxicity and anemia
- Pain at the site of injection

Polymixin B & Colistin

- The polymyxins, discovered in 1947,
- Produced by various strains of *Bacillus polymyxa*.
- *Colistin* is produced by *Bacillus (Aerobacillus) colistinus*.
- Are cationic detergents
- are relatively simple, basic peptides with molecular masses of about 1000 daltons.

Antibacterial Activity and Mechanism of Action

- Activity is restricted to **Gm(-)** bacteria, including *Enterobacter*, *E. coli*, *Klebsiella*, *Salmonella*, *Pasteurella*, *Bordetella*, *Shigella*, most strains of *P. aeruginosa*.
- *Proteus* spp. are intrinsically resistant.
- Polymyxins are surface-active, amphipathic agents. They **interact strongly with phospholipids** and **disrupt the structure of cell membranes**.
- Sensitivity to polymyxin B apparently is related to the phospholipid content of the cell wall-membrane complex.
- The cell walls of certain resistant bacteria may prevent access of the drug to the cell membrane.

➤ Pharmacokinetics

- Both are not absorbed when given orally and are poorly absorbed from mucous membranes and the surfaces of large burns.
- Cleared renally (dose adjustment in renal impairment)

➤ Therapeutic Uses

- Colistin may be useful as a salvage regimen for treatment of infections caused by MDR organisms such as *Stenotrophomonas maltophilia*, *Acinetobacter* spp., or *P. aeruginosa*,
- Infections of the skin, mucous membranes, eye, and ear due to sensitive microorganisms

Untoward Effects

- Muscle weakness and apnea (interference with neurotransmission at the neuromuscular junction)
- Vertigo, and slurred speech.
- Polymyxins are **nephrotoxic**, and administration with aminoglycosides should be avoided if possible

ANTITUBERCULOSIS DRUGS

- Characteristics of mycobacteria that make the diseases chronic & necessitate prolonged treatment
 - Mycobacteria grow slowly and may be dormant in the host for long periods
 - Many antibacterials do not penetrate mycobacterial cell walls, and a portion reside inside macrophages
 - Mycobacteria are agile in developing resistance to single chemotherapeutic agents
- Consequence $\Rightarrow$ effective therapy requires a prolonged course of multiple drugs
 - Patient compliance, drug toxicity, drug interactions, concurrence of other diseases

Tuberculosis

- Two species of Mycobacterium cause tuberculosis:
M. tuberculosis and *M. bovis*
- *M. tuberculosis* is transmitted by inhalation of infective droplets coughed or sneezed into the air by a patient with tuberculosis.
- *M. bovis* is transmitted by milk from diseased cows and first produces intestinal or tonsillar lesions.

Drugs for Tuberculosis

➤ *First-line drugs are superior in efficacy and possess an acceptable degree of toxicity.*

- Include isoniazid, rifampin, pyrazinamide, ethambutol, and **streptomycin (Controversial)**
- Most patients with tuberculosis can be treated successfully with these drugs.

➤ *Second-line drugs are more toxic and less effective*

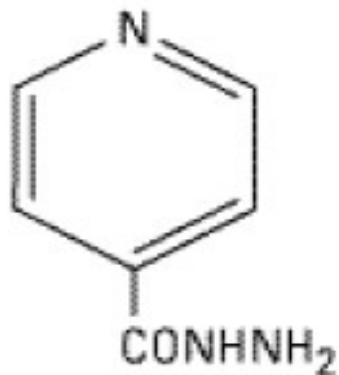
- Indicated only when the *M. tuberculosis organisms* are resistant to the first-line agents.
- Include cycloserine, ethionamide, aminosalicylic acid, rifabutin, quinolones, capreomycin, viomycin, and thiacetazone.

First-line drugs for tuberculosis

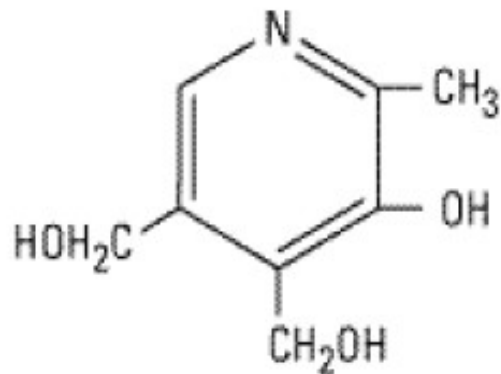
Isoniazid (isonicotinic acid hydrazide, INH)

➤ Chemistry

- Isoniazid is the hydrazide of isonicotinic acid
- Structurally similar to pyridoxine
- A small, simple molecule freely soluble in water



Isoniazid



Pyridoxine

Antibacterial Activity

- The most active drug for the treatment of tuberculosis caused by susceptible strains
- **Is bacteriostatic for "resting" bacilli, but is bactericidal for rapidly dividing microorganisms**
- Remarkably selective for mycobacteria
 - Minimum tuberculostatic concentration = 0.025 to 0.05 mg/ml
 - The growth of other microorganisms is inhibited with concentrations in excess of 500 mg/ml
- Penetrates cells with ease and is just as effective against bacilli growing within cells as it is against those growing in culture media (**Effective intracellular activity**)

Mechanism of Action

- Mycobacterial **catalase-peroxidase** (*katG* encoded) converts isoniazid into an active metabolite
- A primary action of isoniazid is to **inhibit the biosynthesis of mycolic acids (MA)**
- **Resistance can be mapped to mutations in at least five different genes: *katG*, *inhA*, *ahpC*, *kasA*, and *ndh*.**
- Catalase-peroxidase-activated isoniazid binds enoyl-ACP reductase of fatty acid synthase II, which converts Δ^2 -unsaturated fatty acids to saturated fatty acids in the mycolic acid biosynthetic pathway.
- It also inhibits mycobacterial catalase-peroxidase, which may increase the likelihood of damage to the mycobacteria from reactive oxygen species and H_2O_2

Bacterial Resistance

- Mutations in catalase-peroxidase (*katg*) (most common).
- Mutation in the mycobacterial *inhA* and *KasA* genes involved in mycolic acid biosynthesis
- Mutations in NADH dehydrogenase (*ndh*)
- Cross-resistance between isoniazid and other antituberculosis agents (except **ethionamide**) does not occur.
- Approximately 1 in 10^6 tubercle bacilli are genetically resistant to isoniazid

Pharmacokinetics

- Water soluble, well absorbed from oral & parenteral sites
- Does not bind to serum proteins, diffuses readily into all body fluids and cells
 - Caseous tuberculous lesions.
 - Pleural and ascitic fluids, saliva and skin
- Concentrations in CNS & CSF are similar to those in plasma during meningeal inflammation
- INH is acetylated to acetyl INH by *N-acetyltransferase*
 - Genetic polymorphism in the rate of acetylation.
 - ✓ Slow or rapid acetylation is rarely important clinically, although slow inactivators tend to develop peripheral neuropathy more readily.

Clinical Uses

- Isoniazid is the safest and most active mycobactericidal
- Considered the primary drug for use in all therapeutic and prophylactic regimens for susceptible tuberculosis infections
- As a single agent for treatment of latent tuberculosis, which is usually determined by a positive tuberculin skin test.

Adverse Reactions

➤ Allergic Reactions

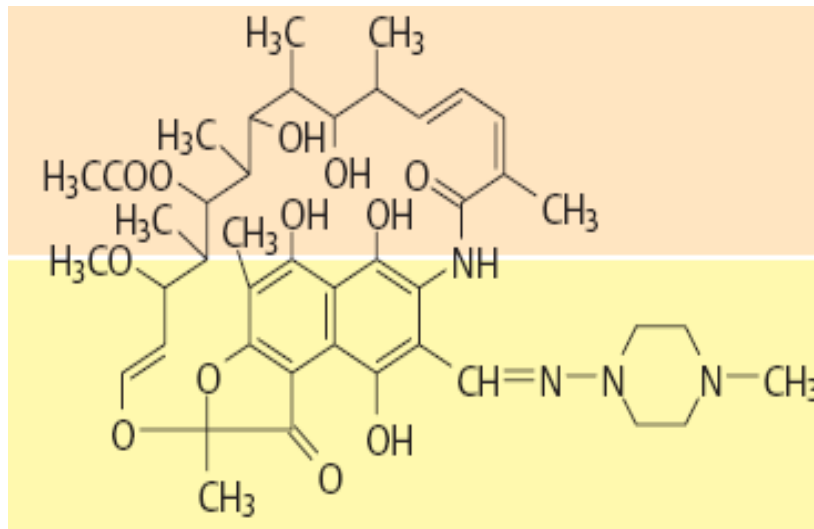
- Fever, skin rashes, Drug-induced systemic lupus erythematosus

➤ Direct Toxicity

- Hepatitis (greater risk in alcoholics & pregnancy and is a contraindication to further use of the drug).
- **Peripheral neuropathy** due to a relative pyridoxine deficiency (more likely in slow acetylators, malnutrition, alcoholism, diabetes, AIDS)
 - ✓ Minimized by administration of **pyridoxine**
- CNS toxicity (memory loss, psychosis, and seizures)
 - ✓ May respond to **pyridoxine**

Rifampin and other rifamycins

- The rifamycins (rifampin, rifabutin, rifapentine) are a group of structurally similar, complex macrocyclic antibiotics produced by *Ammycolatopsis mediterranei*
- Rifampin is a semisynthetic derivative of rifamycin B.
- Rifampin is soluble in organic solvents and in water at acidic pH.



Antibacterial Activity

- Rifampin inhibits the growth of *M. tuberculosis in vitro*.
- *It also inhibits the growth of* nontuberculous mycobacteria
- Rifampin inhibits the growth of most Gm(+) bacteria as well as many Gm(-) microorganisms such as *E. coli*, *Pseudomonas*, *Proteus*, and *Klebsiella*.

Mechanism of Action

- Rifampin binds DNA-dependent RNA polymerase and forms stable drug-enzyme complex, leading to suppression of initiation of chain formation (but not chain elongation) in RNA synthesis.
- Nuclear eukaryotic RNA polymerases are unaffected
- High concentrations can inhibit RNA synthesis in mammalian mitochondria, viral DNA-dependent RNA polymerases, and reverse transcriptases.
- **Bactericidal for both intracellular and extracellular microorganisms.**

Bacterial Resistance

- Occurrence: 1 in 10^7 to 10^8 tubercle bacilli
- Due to an alteration of the DNA-dependent RNA polymerase,
 - Mutations between codons 507 and 533 of the polymerase *rpoB* gene
 - ✓ Reduced binding of the drug to the polymerase
 - ✓ It should not be used alone since resistance develops very fast

Pharmacokinetics

- Well absorbed after **oral** administration and excreted mainly through the liver into bile.
- **Undergoes enterohepatic recirculation**
- Intestinal reabsorption is **reduced** by **food**
- Small amount excreted in urine
- **Distributed widely** in body fluids and tissues
- Rifampin is relatively **highly protein-bound**,
- **Adequate CSF** concentrations are achieved only in the presence of meningeal inflammation

Clinical Uses

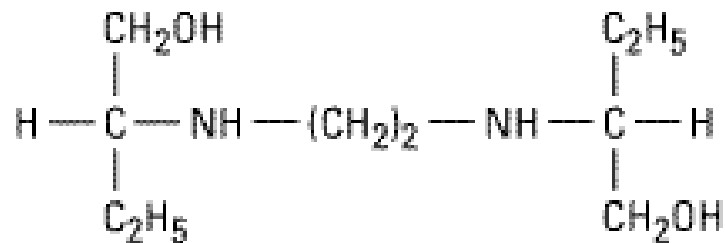
- A first-line antitubercular drug used in the treatment of all forms of **pulmonary** and **extrapulmonary TB**
- Alternative to INH in the treatment of latent TB infection
- In combination with antileprosy agent for the treatment of leprosy
- *Also used in **MRSA** infections*

Adverse Reactions

- Imparts orange color to urine, sweat, tears
- Rashes, thrombocytopenia, and nephritis, cholestatic jaundice and occasionally hepatitis
- It is a strong inducer of most cytochrome P450 isoforms
 - Increases the elimination of numerous drugs including methadone, anticoagulants, some anticonvulsants, protease inhibitors, and contraceptives

Ethambutol

- Ethambutol is a synthetic, water-soluble, heat-stable compound



Ethambutol

Antibacterial Activity

- Nearly all strains of *M. tuberculosis* and *M. kansasii* as well as a number of strains of MAC are sensitive
- Ethambutol has no effect on other bacteria.
- **It suppresses the growth of most INH-and streptomycin-resistant tubercle bacilli.**
- Resistance to ethambutol develops very slowly *in vitro*.

Mechanism of Action and Resistance

- Inhibits **arabinosyl transferases** involved in cell wall biosynthesis.
- The enzyme is involved in **the polymerization reaction of arabinoglycan**, an essential component of the mycobacterial cell wall.
- Bacterial resistance to the drug develops *via* single amino acid mutations in the *embA* gene when ethambutol is given in the absence of other effective agents

Pharmacokinetics

- Well absorbed from the GIT
- About 20% of the drug is excreted in feces and 50% in urine in unchanged form (dose adjustment in renal impairment)
- Is widely distributed in all body fluids
 - It crosses the BBB only if the meninges are inflamed.

Clinical Uses

- Has replaced aminosalicylic acid as a first-line antitubercular drug.
- Commonly included as a 4th drug, along with INH, pyrazinamide, and rifampin, in patients infected with MDR strains.
- Also used in combination in the treatment of *M. aviumintracellulare* infection in AIDS patients.

Adverse Reactions

- Hypersensitivity, loss of visual acuity and **red-green color** blindness (most serious side-effects)
- Ethambutol is relatively contraindicated in children too young to permit assessment of visual acuity and red-green color discrimination.

Streptomycin

- An aminoglycoside antibiotic and was the first drug to reduce tuberculosis mortality.
- It is **bactericidal** against *M. Tuberculosis* in vitro but is **inactive** against **intracellular** organisms.
- Most *M. tuberculosis* strains and nontuberculosis species, such as *M. kansasii* and *M. Aviumintracellulare*, are sensitive.

Antibacterial Activity

- Bactericidal *in vitro*.
- Suppressive activity *in vivo*
- Most *M. tuberculosis* strains and nontuberculosis species, such as *M. kansasii* and *M. aviumintracellulare*, are sensitive
- **It penetrates into cells poorly, and is active mainly against extracellular tubercle bacilli**

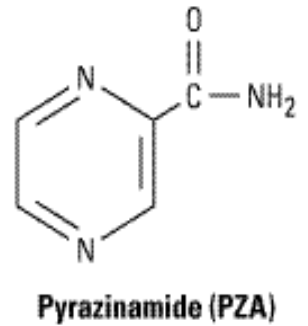
Mechanism of action and resistance

- Prokaryotic **ribosomal protein synthesis inhibitor**
- Spontaneous resistance occurs in approximately 1 in 10^6 tubercle bacilli,
- Resistance is related to a point mutation that involves the gene (rpsl or rrs) that encodes for **ribosomal proteins** and binding sites
- About 80% of strains that are resistant to INH and rifampin are also resistant to streptomycin.

Clinical Use in TB

- As a fourth drug in combination with INH, rifampin, and pyrazinamide in patients at high risk for drug resistance
- Employed when an injectable drug is needed or desirable, (severe, possibly life-threatening forms of TB, and in treatment of infections resistant to other drugs)
- Other drugs are always given simultaneously to prevent emergence of resistance

Pyrazinamide



- Synthetic pyrazine analog of nicotinamide
- It is stable, slightly soluble in water, and quite inexpensive.
- **At neutral pH, it is inactive in vitro, but at pH 5.5 it inhibits tubercle bacilli**
- Drug is taken up by macrophages and exerts its activity against **intracellular** organisms residing within acidic environment.

Mechanism of Action and Resistance

- Exact MOA unknown, although its target appears to be the mycobacterial fatty acid synthetase involved in mycolic acid biosynthesis.
- It is a prodrug activated to the active pyrazinoic acid by the mycobacterial enzyme pyrazinamidase
- Mutation in the gene (*pncA*) that encodes pyrazinamidase is responsible for drug resistance
- Resistance develops fairly readily, but there is no cross-resistance with INH or other antimycobacterial drugs.

Pharmacokinetics

➤ Pharmacokinetics

- Well absorbed from the GIT and widely distributed in body tissues, including inflamed meninges

➤ Clinical uses

- An important front-line drug used in conjunction with INH and rifampin in short-course (ie, 6-month) regimens as a "**sterilizing**" agent active against residual intracellular organisms that may cause relapse.

Adverse effects

- **Hepatotoxicity** (in 1–5%), nausea, vomiting, drug fever, and hyperuricemia (occurs uniformly and may provoke acute gouty arthritis)

- **Category I**

- It includes those new patients who have smear-positive Pulmonary TB and those who seriously ill patients with smear-negative Pulmonary and Extra-pulmonary TB..
- The treatment regimen for this category is 2 (RHZE) / 6 (EH) or OR 2 (RHZE) / 4RH

- **Category II**

- This category is applied to a group of TB patients:
 - Who relapsed after being treated and declared free from the disease, OR

Please refer the TB guide line for dosages and regimen

- In those patients who are previously treated for more than one month with SCC or LCC, and found to be smear positive upon return, OR
- Who still remain smear positive while under treatment ,at month five and beyond
- The treatment regimen for this category is: 2 SE (RHZ) / 1E (RHZ) / 5 E3 (RH)3.

- **Category III**

- This refers to patients who have smear negative Pulmonary TB, Extra-pulmonary TB and TB in Children.
- The regimen consists of 8 weeks of treatment with, Ethambutol, Rifampicin, Isoniazid and Pyrazinamide during the intensive phase followed by Ethambutol and Isoniazid six months [2(RHZE/)/6(EH)]

- **Category IV**

- Treatment of chronic cases
- those cases that continue to be smear-positive after completion of a fully supervised (initial phase and continuation phase) treatment with the treatment regimen

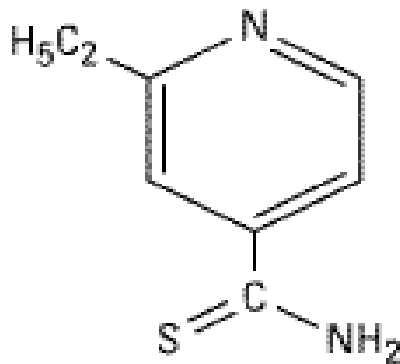
Alternative Second-Line Drugs for Tuberculosis

Quinolones

- The fluoroquinolones are highly active against *M. tb* as well as nontuberculous mycobacteria
- important components of treatment regimens of MDR TB
- C-8-methoxy-fluoroquinolones (gatifloxacin and moxifloxacin) are the most active
- Cross-resistance within the entire class is common
- To prevent development of resistance, the most active fluoroquinolones should be used, and only in combination with other antimycobacterial agents

Ethionamide

- *A derivative of isonicotinic acid* and is chemically related to isoniazid.
- A secondary agent used in combination when primary agents are ineffective or contraindicated
- It is a **bacteriostatic antituberculosis agent**.



Ethionamide

➤ Mechanism of Action

- EtaA, an NADPH-specific, FAD-containing monooxygenase, converts ethionamide to a sulfoxide, and then to 2-ethyl-4-aminopyridine
- Inhibits the activity of the *inhA* gene product, the enoyl-ACP reductase of fatty acid synthase II
 - ✓ Inhibition of mycolic acid biosynthesis and consequent impairment of cell-wall synthesis.

➤ ***Pharmacokinetics***

- Well absorbed following oral administration
- Rapidly and widely distributed to all body tissues and fluids, including the cerebrospinal fluid.
- Cleared by hepatic metabolism; *it inhibits the acetylation of isoniazid in vitro.*
- Less than 1% of the drug is eliminated in the urine unchanged.

➤ Therapeutic Uses

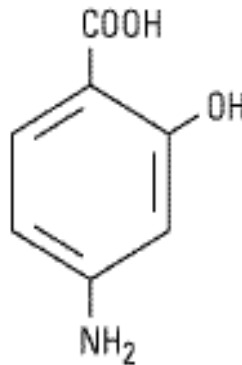
- A secondary agent, used concurrently with other drugs only when therapy with primary agents is ineffective or contraindicated.

➤ Untoward Effects

- Anorexia, nausea and vomiting, gastric irritation, and a variety of **neurologic symptoms**. Severe postural hypotension, mental depression, drowsiness, convulsions, peripheral neuropathy.
- **Pyridoxine** (vitamin B₆) relieves the neurologic symptoms
- Metallic taste

P-Aminosalicylic acid

- Folate synthesis antagonist that is active almost exclusively against *M. tb*.
- *It is structurally similar to p-aminobenzoic acid (PABA) and to the sulfonamides*



Aminosalicyclic acid (PAS)

➤ Mechanism of Action

- It is a **folate synthesis antagonist** that interferes with the incorporation of **PABA** into folic acid.
- PAS is **bacteriostatic**

➤ *Therapeutic Uses*

- A second-line agent

➤ *Untoward Effects*

- GI problems(anorexia, nausea, epigastric pain, abdominal distress, and diarrhea)
- Patients with peptic ulcers tolerate the drug poorly
- Hypersensitivity reactions
- Hematological abnormalities (leukopenia, agranulocytosis, eosinophilia, lymphocytosis, thrombocytopenia).

Cycloserine

- Cycloserine is a broad-spectrum antibiotic produced by *Streptomyces orchidaceus*.
- *It is structural analogue of D-alanine*
- ***Antibacterial Activity***
 - No cross-resistance between cycloserine and other tuberculostatic agents.
 - D-alanine blocks the antibacterial activity of cycloserine
 - Inhibitory to *M. tb* and active against *E. coli*, *S. aureus*, and *Enterococcus*, *Nocardia*, and *Chlamydia spp.*

➤ Mechanism of Action

- Cycloserine and D-alanine are structural analogs; thus, cycloserine inhibits reactions in which D-alanine is involved in bacterial cell-wall synthesis

➤ *Pharmacokinetics*

- Rapidly absorbed When given orally (70% to 90%)
- Distributed throughout body fluids and tissues
- CSF concentrations are approximately the same as those in plasma
- About 65% of a parenteral dose is excreted unchanged in the urine in the first 72 hours.

➤ Therapeutic Uses

- In the treatment of **MDR tuberculosis** and is useful in renal tuberculosis, since most of the drug is excreted unchanged in the urine

➤ *Untoward Effects*

- CNS (headache, tremor, vertigo, confusion, nervousness, irritability, psychotic states with suicidal tendencies, paranoid reactions, catatonic and depressed reactions, visual disturbances, seizures)
- Contraindicated in individuals with a history of **epilepsy** and should be used with caution in individuals with a history of depression, as suicide is a risk.

Other drugs

➤ Kanamycin & Amikacin

- Aminoglycoside antibiotics
- Amikacin is indicated for treatment of tb suspected or known to be caused by streptomycin-resistant or MDR strains

➤ *Capreomycin*

- *A polypeptide antibiotic derived from *Streptomyces capreolus*.*
- *It is **bacteriostatic** against most strains of *M. tuberculosis*, including the MDR strain.*
- Used as a second-line agent in combination with other drugs.
Particularly useful in multidrug regimens for the treatment of drug resistant tb (esp with streptomycin resistance)
- Capreomycin is associated with ototoxicity and nephrotoxicity